

RESEARCH ARTICLE



WILEY

Artificial intelligence for Sustainable Development Goals: Bibliometric patterns and concept evolution trajectories

Aakash Singh¹ | Anurag Kanaujia¹ | Vivek Kumar Singh¹ | Ricardo Vinuesa²

¹Department of Computer Science, Banaras Hindu University, Varanasi, India

²FLOW, Engineering Mechanics, KTH Royal Institute of Technology, Stockholm, Sweden

Correspondence

Vivek Kumar Singh, Department of Computer Science, Banaras Hindu University, Varanasi 221005, India.

Email: vivek@bhu.ac.in

Ricardo Vinuesa, FLOW, Engineering Mechanics, KTH Royal Institute of Technology, Stockholm SE-10044, Sweden.

Email: rvinuesa@mech.kth.se

Funding information

HPE Aruba Centre for Research in Information Systems, Grant/Award Number: M-22-69; Science and Engineering Research Board, Grant/Award Number: MTR/2020/000625

[Correction added on 3 August 2023, after first online publication: Corresponding author's e-mail address has been corrected in this version.]

Abstract

The development of artificial intelligence (AI) as a field has impacted almost all aspects of human life. More recently it has found a role in addressing developmental challenges, specifically the Sustainable Development Goals (SDGs). However, there are not enough systematic studies on analysis of the role of AI research towards the SDGs. Therefore, this article attempts to bridge this gap by identifying the major bibliometric trends and concept-evolution trajectories in the area of AI applications for sustainable-development goals. The research publication data for the last 20 years in the areas of artificial intelligence, machine learning, deep learning, and so forth, is obtained and computationally analysed using a framework comprising bibliometrics, path analysis and content analysis. The findings show an incremental trend in overall publications on the application of AI for SDGs across the different regions of the world. SDGs 3 (good health & well-being) and 7 (affordable and clean energy) are found as the areas with the most applications of AI. In SDG3, the literature reflects application of AI techniques such as deep learning for precision and personalised medicine while in SDG7, a number of studies have employed AI techniques for the integration of systems for efficient generation of solar power and improving the energy efficiency of a building. Furthermore, SDG 4 (quality education), SDG 13 (climate action), SDG 11 (sustainable cities and communities) and SDG 16 (peace, justice and strong institutions) are the other SDGs where AI approaches and techniques are applied. The analytical results present a detailed insight of application of AI for achieving the SDGs.

KEYWORDS

artificial intelligence (AI), bibliometrics, path analysis, Sustainable Development Goals (SDGs)

1 | INTRODUCTION

The development of artificial intelligence (AI) as a field has been all encompassing. It has impacted almost all aspects of human life, made possible unprecedented developments in areas of communication, medicine, engineering, sustainability, and so forth, (Allen et al., 2020).

AI broadly refers to the design of intelligent machines that enable high-level cognitive processes like thinking, perceiving, learning, problem solving and decision making. This field has gained broader attention with significant advances in data collection and aggregation, analytics and availability of suitable computer processing power (National Strategy for Artificial Intelligence, 2018). AI has not

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2023 The Authors. *Sustainable Development* published by ERP Environment and John Wiley & Sons Ltd.

only attracted the attention of the academic research community but also has become a focus area for technology and service companies, playing a major role in the *Fourth Industrial Revolution* (Moavenzadeh, 2015). AI systems can help people acquire new skills, democratise services, design and deliver faster production times and quicker iteration cycles, reduce energy usage, provide real-time environmental monitoring for pollution and air quality, enhance cybersecurity defences, reduce healthcare inefficiencies, create new kinds of recreational experiences, and improve real-time translation services to connect people around the world. Looking at the wide applicability and impact of AI, several national governments have opted to promote and regulate AI adoption in various sectors (Baum et al., 2023), through national plans and strategies. For instance, UK (National AI Strategy¹), Japan (AI Strategy 2019 & Integrated Innovation Strategy 2020²), China (New Generation AI Development Plan for 2030³), India (National Strategy for Artificial Intelligence (NSAI)), UAE (UAE Strategy for Artificial Intelligence (AI)⁴), South Korea (Toward AI World Leader beyond IT⁵), Canada (Pan-Canadian AI Strategy⁶), and so forth. are some of the countries which have led the effort towards development of systematic plans for the wider adoption of AI in governance and other sectors.

The Sustainable Development Goals (SDGs) are a set of 17 guiding targets adopted by United Nation, each focusing on an aspect of human development and sustainability of ecosystems. The constitution and adoption of these goals signifies a universal call to action to address the urgent environmental, political and economic challenges being faced by the world (UNGA, 2015⁷; United Nations, 2020⁸). These SDGs have played an important role in promoting research and development in novel technologies since their adoption by the United Nations in 2015. As an activity, *sustainable development* has been defined as 'development that meets the needs of the present without compromising the ability of future generations to meet their own needs'. It calls for concerted efforts towards building an inclusive, sustainable and resilient future for the people and the planet (Brundtland Commission, 1987⁹). In order to make considerable progress in achieving the targets of SDGs, the role of science and technology will be central.

AI has emerged as a transformative technology that holds great promise in economic, social, medical, security, and environmental applications, including some of the major important concerns of the current century included under the SDGs. The applications of AI have helped in addressing developmental challenges, specifically sustainable development has received attention in recent times (Vinuesa

et al., 2020). In this regard, the SDGs are an area where practical applications of AI technology have grown rapidly (Goralski & Tan, 2020; Vinuesa et al., 2020). Though AI applications in achieving SDG targets are now being recognised, not much is known about the quantum of research on AI for SDGs, the major focus areas of AI for SDGs, the prominent AI techniques and methods being applied to SDGs, and so forth. Therefore, there exists a research gap in terms of in-depth knowledge about applications of AI for SDGs. This has motivated us to undertake this study. This article attempts to address this research gap by using analytical methods from bibliometrics and content analysis. More specifically the article attempts to answer the following research questions.

RQ1. What quantum of AI research, and from which regions of the world, is focused on SDGs?

RQ2. Whether different SDGs attract differential attention from AI researchers, as measured in terms of publication and citation activity?

RQ3. What are the major thematic evolution trajectories in AI research for SDGs?

RQ4. Which prominent AI methods and models have been applied to achieve different SDG targets?

In order to answer these questions, the research publications data on AI for SDGs has been explored and analysed using Bibliometric analysis, Path analysis and Content analysis. Bibliometric Analysis is a method, which is applied to explore large amount of literature and present an all-encompassing review of the literature (Donthu et al., 2021). It identifies major bibliometric trends such as most influential publications, authors, subdomains, and so forth, (Farrukh et al., 2020; Kumar et al., 2021). However, a major criticism of this method is that it often ignores more detailed trends. Therefore, to uncover more meaningful information about research growth and thematic structure of the research area, methods of Path Analysis and Content Analysis have been used. Path Analysis involves exploring the major knowledge flow paths through tracing the citations of important publications (Lathabai et al., 2018; Liu & Lu, 2012). Content Analysis utilises the insights from the path analysis and involves in depth review of most important publications identified during the path analysis. The methodology have been described further in the data and methods section.

1.1 | Major contributions

This study attempts to address the research gap on more detailed information on AI applications for SDGs. This study is among the first of its kind and makes following major contributions. It quantifies the total research activity on AI for SDGs from different parts of the world, how has it changed over time, which regions of the world

¹National AI Strategy – <https://www.gov.uk/government/organisations/office-for-artificial-intelligence>.

²AI Strategy 2019 & Integrated Innovation Strategy 2020 –<https://www.meti.go.jp/press/2020/01/20210115003/20210115003-3.pdf>.

³New Generation AI Development Plan for 2030 – <https://www.theconstructsim.com/98-chinas-ai-plan-for-2030/>.

⁴UAE Strategy for Artificial Intelligence (AI) – <https://ai.gov.ae/>.

⁵Towards AI World Leader beyond IT – <https://oecd.ai/dashboards/countries/SouthKorea>.

⁶Pan-Canadian AI Strategy – <https://cifar.ca/ai/>.

⁷<https://upload.wikimedia.org/wikipedia/commons/d/d5/N1529189.pdf>.

⁸<https://sdgs.un.org/publications/sustainable-development-goals-report-2020-24686>.

⁹<https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf>.

are working towards AI applications to SDGs and how do they collaborate with each other for this purpose. In addition, the knowledge flows in AI research for SDGs has been explored to identify major application areas along with AI methods and models applied to address different SDG targets. Thus, the study provides a detailed insight into applications of AI research for SDGs by underlining its application areas, major themes, most important SDG targets, and major methods and models applied.

1.2 | Organisation of the paper

This article is organised as follows. The explanation of AI, its impacts, SDGs, their connection with technology, and research questions for this study are listed in the introduction section. Section 2 surveys some related work on the theme. The data and methodology are detailed in Section 3. Section 4 with descriptive results presents the observations from bibliometric path analysis and content analysis. Section 5 of the article presents in-depth discussion on the main findings including SDG wise summaries of research trends in AI research for SDGs. The practical implications of study are highlighted in section 6. The conclusion section of the article underlines the relevance of the findings and suggests possible future work related to this study.

2 | RELATED WORK

The research community has recently started to explore the applications of AI in SDGs, and this new confluence of two areas has emerged. Among these studies the common theme relates to: (a) evaluating the impact of existing techniques of AI on achievement of SDGs, and (b) regulating the application of AI on SDGs. Early on, Chui et al. (2018) looked at the role of AI for social good by compiling about 160 cases where AI had made a social-impact. While the impact of AI on individual areas of sustainability have been studied by several others, the other notable study covering all SDGs was reported by Vinuesa et al. (2020). It used an expert-elicitation approach to estimate the role of AI in achieving the SDGs and suggested that AI can enable the accomplishment of 134 targets, while it may also inhibit the progress on 59 targets. In contrast, Yeh et al. (2021), conducted a public perception survey about the connection between the two areas. Another such study categorises SDGs into five groups on the basis of the type of impact of AI on them (Sætra, 2021). It specifically brings forth the potential negative impacts of AI which could result from the dominance of large firms and selected nations on the governance structure of AI applications. Gupta et al. (2021) have conducted a more detailed study of AI applications at an SDG indicator level.

Another study has explored the impact of AI research on different domains where AI applications can facilitate social good (Wamba et al., 2021). Goralski and Tan (2020) have looked at three case studies (water, agriculture and sanitation & health) on the applications of AI to improve the efficiency and efficacy of resource management.

They have compared AI to a double-edged sword, which must be used carefully by the relevant stakeholders. This is particularly relevant when using AI for sustainability applications, which requires the development of interpretable models (Vinuesa & Sirmacek, 2021), or in complex social contexts such as that of pandemics (Vinuesa et al., 2020b). Some region-specific study has looked at the relationship between AI and sustainable development in China (Liengpunsakul, 2021) and in India (Singh et al., 2022). These studies have used approaches favouring a qualitative analysis of available subject expertise and literature, and we could find only a few very recent previous articles on quantitative analysis, namely, Wamba et al., 2021 and Liu et al., 2023. A new platform named AI4SDGs Think Tank¹⁰ has started which evaluates AI projects for their positive or negative impact on SDGs.

There are, however, no significant accounts of available literature on the topic of AI for SDGs, in terms of the quantum of research on AI for SDGs, the major focus areas of AI for SDGs, the prominent AI techniques and methods being applied to SDGs, and so forth. Therefore, there exists a research gap in terms of in-depth knowledge about applications of AI for SDGs. This article attempts to address this research gap by using analytical methods from bibliometrics, path analysis and content analysis.

3 | DATA AND METHODOLOGY

The study is based upon the publication metadata catalogued in Dimensions¹¹ database from 2001 to 2020. Dimensions database provides an automated classification of publications into various SDGs. The publication metadata was accessed by the means of a dedicated API granted from the database. The query to obtain the data was as follows:

```
'search publications in title_abstract_only for
"\Artificial intelligence\" OR \"machine learning\" OR
\"Deep learning\" where category_sdg in ["40001",
"40002", "40003", "40004", "40005", "40006",
"40007", "40008", "40009", "40010", "40011",
"40012", "40013", "40014", "40015", "40016",
"40017"] and year = '+year+' and type = "article"
return publications [id + doi + title + abstract +
category_for + category_sdg + authors + authors_
count + journal + year + type + issn + research_orgs
+ research_org_countries + concepts + concepts_
scores + times_cited+ funders + funder_countries +
referenced_pubs + publisher + researchers +date +
pages + open_access_categories_v2]'
```

This query yielded a total of 20,511 documents which were then imported to the working database. Dimensions database was chosen

¹⁰<https://ai-for-sdgs.academy/about>.

¹¹Dimensions database is available at www.dimensions.ai.

over other available databases considering its wider coverage (Singh et al., 2021) and its automated classification of publications into various SDGs. The metadata downloaded consisted of twenty-five (25) attributes for each publication, these include publication id (unique identification number assigned to each publication), DOI, title of the publication, abstract, disciplinary “category” assigned by Dimensions, SDG category, that is, the SDG(s) to which the publication has been associated by Dimensions’ AI algorithm, author(s) of the publication, number of authors, name of the journal in which the publication appeared, year of publication, type of publication, ISSN, author affiliation, country in which author’s organisation is located, concepts associated with the publication, concept scores for the associated concepts, number of citations received by the publication, references cited in the papers, and so forth. This data was retrieved in August 2021 and analysed using methods of standard bibliometric analysis, Path Analysis and a Content-based analysis.

3.1 | Bibliometric analysis

Bibliometric Analysis is a method which is applied to explore large amount of literature and can identify major trends such as most influential publications, authors, subdomains, collaborations, and so forth, (Donthu et al., 2021; Farrukh et al., 2020; Kumar et al., 2021). For carrying out bibliometric analysis, a standard bibliometrics approach was used. From the publication metadata, first of all, the total number of publications in AI for SDGs were identified. Thereafter, major countries/ regions contributing to AI for SDGs research were identified by processing author affiliation field of the metadata. Country/ region level collaboration networks were also drawn to understand the scientific collaboration structure of AI for SDGs research. The major journals publishing the AI for SDGs research are identified next by processing the publication source field of the publication metadata. Subsequently, the research output in each SDG is measured and a map of contribution of different countries in different SDGs is created.

After looking at the overall picture of the AI for SDGs research output, the SDGs in which most research output is found are analysed in detail. For each such SDG, the subject areas from which research output is coming is identified. The key concepts occurring in publications for each SDG are identified and plotted in a density plot to understand the thematic structure of research publications in each SDG. The major concept evolution trajectories are identified and analysed next by following the path analysis methodology.

3.2 | Path analysis

This study also utilised the network scientometrics approach (path analysis) to observe the development in the literature of AI for SDGs from 2001 through 2020. For this purpose, the extracted data was cleaned to remove records that did not have corresponding values for ‘referenced_pubs’ and used to create the citation network

through publication details and cited reference information. This network is further analysed using the approach described by Lathabai et al. (2018).

Complex network analysis offers a multitude of methods and tools for addressing wide range of problems in real-world systems that can be modelled as networks. Scientific and technological (patent) literature is two such systems that consists of several relationship between entities such as works (papers or patents), authors/inventors, institutional affiliations/assignees, and so forth. One of the major relationships is citations, where for a pair of works, the latest work cites previous related work. This results in the formation of scientific citations network or citation network of scholarly publications/papers (De Solla Price, 1965; Garfield et al., 1964) and patent citation networks or citation networks of patents. Among the different analyses possible in networks, the path analysis helps to identify the important evolutionary trajectories or connective threads of knowledge (Hummon & Dereian, 1989). Path analysis has been used by many for determining important evolutionary paths that might have served as backbone of development of different fields. Few examples are evolutionary trajectory study for medical technologies (Mina et al., 2007), ‘Coronary angioplasty’ (Tampubolon & Ramlogan, 2007), human resource development (Jo et al., 2009) and archaeology (Brughmans, 2013), and so forth.

Citation networks are by default unweighted. Path analysis consists of two major steps – (i) weight assignment or conversion of unweighted network into weighted network and (ii) search or trace through the weighted network to retrieve paths. Original Hummon and Dereian (1989) framework used traversal as the basis of weight assignment of arcs (directed citation links) for the creation of weighted network. A break-through in path analysis literature was brought-in by Batagelj (2003) when computationally efficient traversal-based weight assignment method namely search path count (SPC) method was introduced. Generally, the traversal-based weight assignment schemes can be termed together as SPX schemes. A major development in search scheme happened with the introduction of search schemes like backward search, key-route search, global search, and so forth, with a provision for tolerance. These schemes, especially multiple key-route search, empowered the ability of path analysis to retrieve more important evolutionary trajectories in scientific literature. However, due to the dependency on SPX methods on the global structure of the network, in a connected component, highly (SPX) weighted arcs seem to clutter together, thereby limiting the effectiveness of multiple keyroute search to retrieve multiple paths. In order to address this limitation, (Lathabai et al., 2018) introduced a novel weight assignment method, namely the Flow Vergence (FV) gradient method for weight assignment and also upgrading the Liu-Lu approach (Liu & Lu, 2012). with state-of-the-art integrated framework in path analysis. With the framework of Lathabai et al. (2018), if we use SPX and FV gradient as weight assignment methods, and existing search schemes, following paths can be retrieved: SPX-forward paths, SPX-backward paths, SPX-keyroute paths, SPX-critical paths, FV-forward paths, FV-backward paths, FV-keyroute paths and FV-critical paths. In this work, we use SPC (among SPX) weight assignment and

therefore the retrieved paths will be SPC-forward path, SPC-backward path, SPC-keyroute path, SPC-critical path, FV-forward path, FV-backward path, FV-keyroute path and FV-critical path.

Brief description of SPC and FV gradient weight assignment methods and the search schemes are given in Appendix A.

3.3 | Content analysis

Most of the publications that appear in important paths deal with AI techniques applied to different problems from various fields relating to SDGs. These publications were identified, chronologically arranged and their full texts were accessed from their respective sources. The specific selected characteristics of these studies were brought out by diligent and rigorous analysis of the content of these works by manually going through the abstracts and overall content of these works. The characteristics looked at in these studies were *area(s) of application, theme, SDG target covered, AI method(s)/ models studied/used, nature of the study, and the Inference/Conclusion* drawn by the studies. Although this is a time-consuming and subjective method, it is advantageous if the requisite care is taken during the analysis. Useful insights about popular areas of research, methods used, and so forth, can be drawn by utilising the results of path analysis. For the purpose of this study, articles were assigned one of the five nature (*Predictive (model) Analysis, Descriptive Analysis, Review, Validation Study, Recommendation Analysis*). A chart showing the AI methods/models used in these articles was prepared and is presented as a part of the summary of these qualitative findings. An alternative visualisation of the findings of the content analysis is also presented.

4 | DESCRIPTIVE RESULTS

This section portrays the results obtained from the descriptive analyses that were conducted on the data. These analyses not only helped to visualise the data in a meaningful way but also to formalise the components that needed detailed attention. Four major aspects were focused upon including temporal, regional, collaborative, and quantitative. The temporal analysis includes visualising the trend of publications and their citations over the 20 years, for all the SDGs. The regional analysis revolves around identifying research hotspots in the world based on geotags of each publication. Revelling the collaboration patterns that exist in the domain among authors from different regions was the motive of collaborative analysis. And, the last analysis was used to quantify the research output with the available variables, that is, individual SDGs and the most active regions.

4.1 | Temporal patterns in AI research for SDGs

The total number of publications on AI for SDGs (hereafter referred to as AI4SDG) shows an exponential growth post the adoption of SDGs by the United Nations in 2015 (Figure 1). The overall number

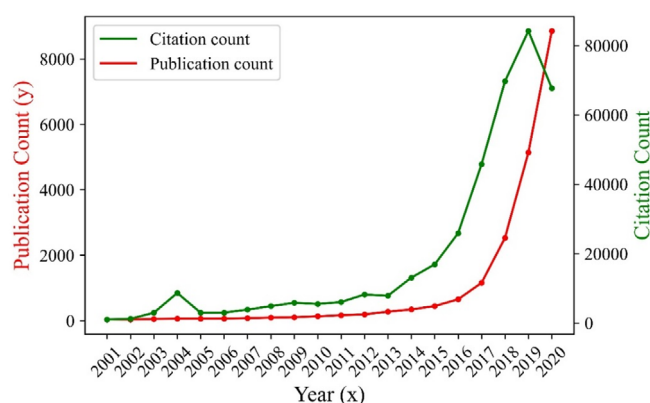


FIGURE 1 The year-wise trend in publication count for papers in AI applications on SDGs. A sharp increase in the number of publications can be seen from 2016 onwards, after the adoption of SDGs in 2015. AI, artificial intelligence; SDGs, Sustainable Development Goals.

of citations received by the publications on AI4SDGs shows a corresponding increase from 2012 to 2015. It may be noted that this time frame is associated with the renaissance of artificial intelligence, in terms of origin of deep learning paradigm in AI which are capable of superior performance in a wide variety of problem domains. AI-based models like “AlexNet” and “Alpha Go” had proved their potential in solving real-world problems (Haenlein & Kaplan, 2019; Muthukrishnan et al., 2020). These new developments may be seen as responsible to have bought back the research community's attention towards AI.

4.2 | Top regions contributing to AI research for SDGs

The most active geographical regions/ countries around the world focusing on AI4SDG research are the United States, Western Europe, China, Japan, Australia, and India (Figure 2). Areas on the east coast of United States (New York, Washington, Hawaii), Canada (Montreal, Ottawa, Toronto, etc.), United Kingdom, Norway, Sweden, France, Germany, Italy, India (New Delhi, Bangalore), Singapore, China (Beijing, Shanghai), Hong Kong, Japan, and Australia (Sydney, Melbourne) show the highest density in the map and would have the greatest number of research publications on AI4SDG. On the other hand, South American countries Mexico, Brazil, Argentina, Africa and parts of Middle East had lower number of publications. In addition to this, the number of publications in different regions were also found to vary by different SDGs. For examples, it was observed that the African region has most of its publications on SDG 2 (Zero Hunger) and SDG 3 (Good Health and Wellbeing). In India and China, the larger percentage of publications were related to SDG 3, SDG 7, and SDG 11. Whereas, the USA, UK, Europe, and Japan had publications relating to all six top-performing SDGs. Among the South American countries, Brazil and Columbia had the largest number of publications. These were mostly related to

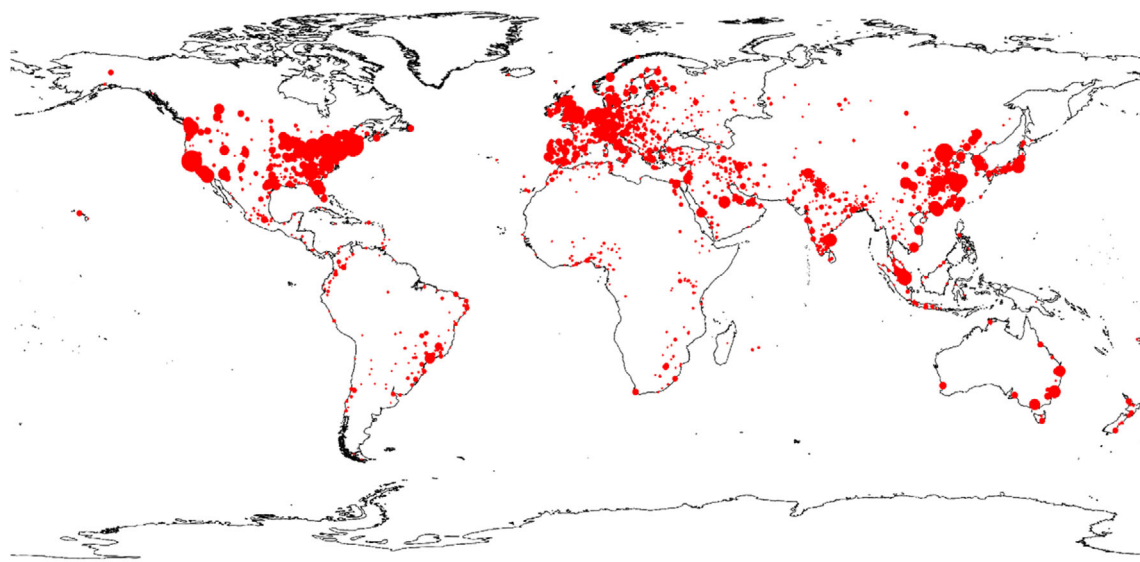
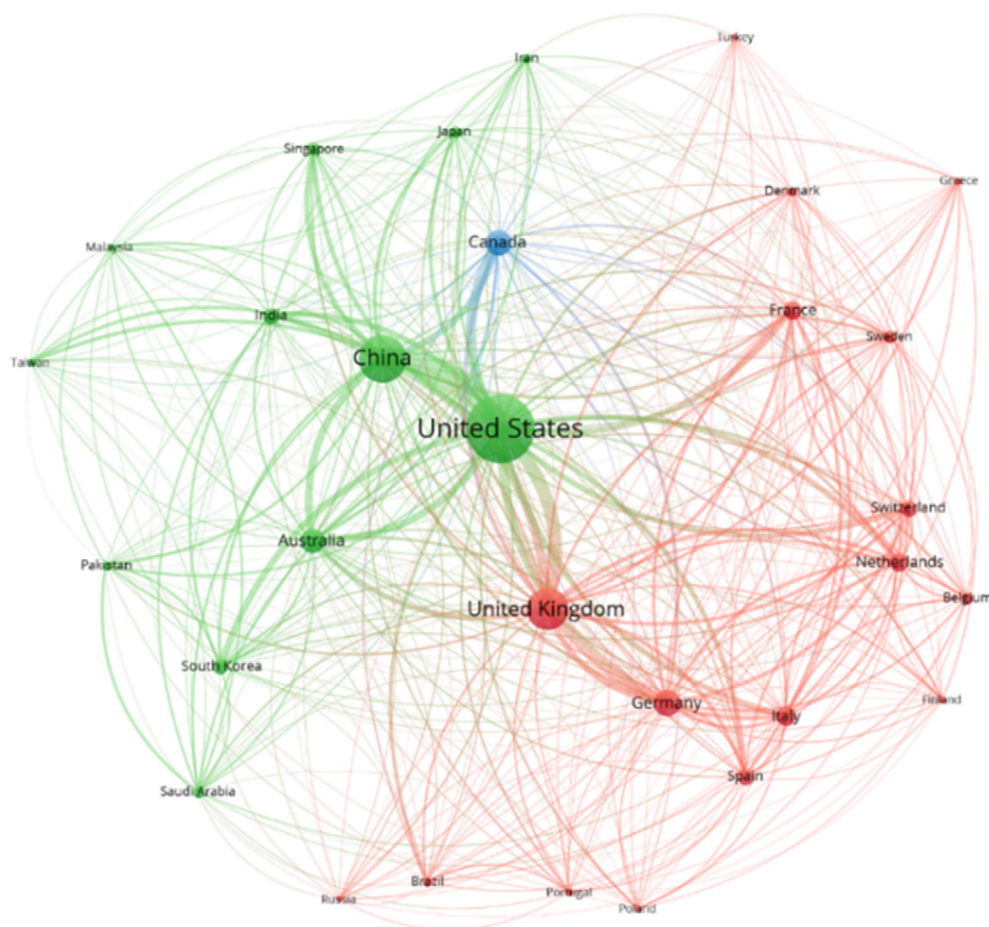


FIGURE 2 Regional distribution of research on AI4SDG. Plotted by using the “location” field of the publication metadata, the size of dot shows the number of publications. The areas with high density of dots had high productivity in AI4SDG research. Geoplot and GeoPandas libraries of python were used to prepare this plot.

FIGURE 3 Region-wise publications and collaborations between different countries are shown by arcs, in AI for SDGs research. AI, artificial intelligence; SDGs, Sustainable Development Goals.



SDG 3 and SDG 7, whereas other SDGs did not have many publications related to AI4SDG from this region. It is observed that research giant countries have been focusing more on SDGs relating to

infrastructure, environment, education, and health and the smaller and underprivileged countries target specifically SDGs relating to health and hunger reflecting their societal needs. This is in

TABLE 1 The top journals publishing papers on applications of artificial intelligence for Sustainable Development Goals.

Journal	Publisher	TP SDG	TC SDG	CPP SDG
IEEE Access	Institute of Electrical and Electronics Engineers (IEEE)	619	8993	14.53
Journal of Physics Conference Series	IOP Publishing	279	440	1.58
Sensors	MDPI	278	5133	18.46
Energies	MDPI	259	3736	14.42
Remote Sensing	MDPI	205	4021	19.61
Applied Sciences	MDPI	198	1933	9.76
PLOS ONE	Public Library of Science (PLOS)	189	4635	24.52
Scientific Reports	Springer Nature	161	4502	27.96
Sustainability	MDPI	148	1589	10.74
Applied Energy	Elsevier	136	7124	52.38
IOP Conference Series Materials Science and Engineering	IOP Publishing	134	145	1.08
International Journal of Innovative Technology and Exploring Engineering	Blue Eyes Intelligence Engineering and Sciences Engineering and Sciences Publication – BEIESP	128	53	0.41
Procedia Computer Science	Elsevier	128	1228	9.59
IEEE Internet of Things Journal	Institute of Electrical and Electronics Engineers (IEEE)	112	3268	29.18
International Journal of Recent Technology and Engineering	Blue Eyes Intelligence Engineering and Sciences Engineering and Sciences Publication – BEIESP	101	69	0.68
Energy	Elsevier	100	2770	27.70
ISPRS – International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences	Copernicus Publications	92	268	2.91
IOP Conference Series Earth and Environmental Science	IOP Publishing	84	90	1.07
IEEE Transactions on Intelligent Transportation Systems	Institute of Electrical and Electronics Engineers (IEEE)	83	1919	23.12
International Journal of Engineering and Advanced Technology	Blue Eyes Intelligence Engineering and Sciences Engineering and Sciences Publication – BEIESP	81	43	0.53

conformity with the observations of Chui et al. (2018), which associated 17 SGDs with wider social and environmental issues.

4.3 | Region-wise collaboration network of AI research for SDGs

The collaborations between different regions were visualised using VoS viewer (Figure 3). The US, The UK, China, Canada, Australia, and Germany had the most publications in collaboration with international authors. The US, the UK and China were the top three closest partners followed by the US and Canada, The UK and Germany, Australia and China, as well as UK and China. It may also be noted that most European countries have close ties with each other when it comes to research collaboration in the domain. These collaboration patterns not only outline the geographical and ideational proximity but also the trends that persist in the transfer of knowledge and technology among these regions.

4.4 | Top journals publishing AI research for SDGs

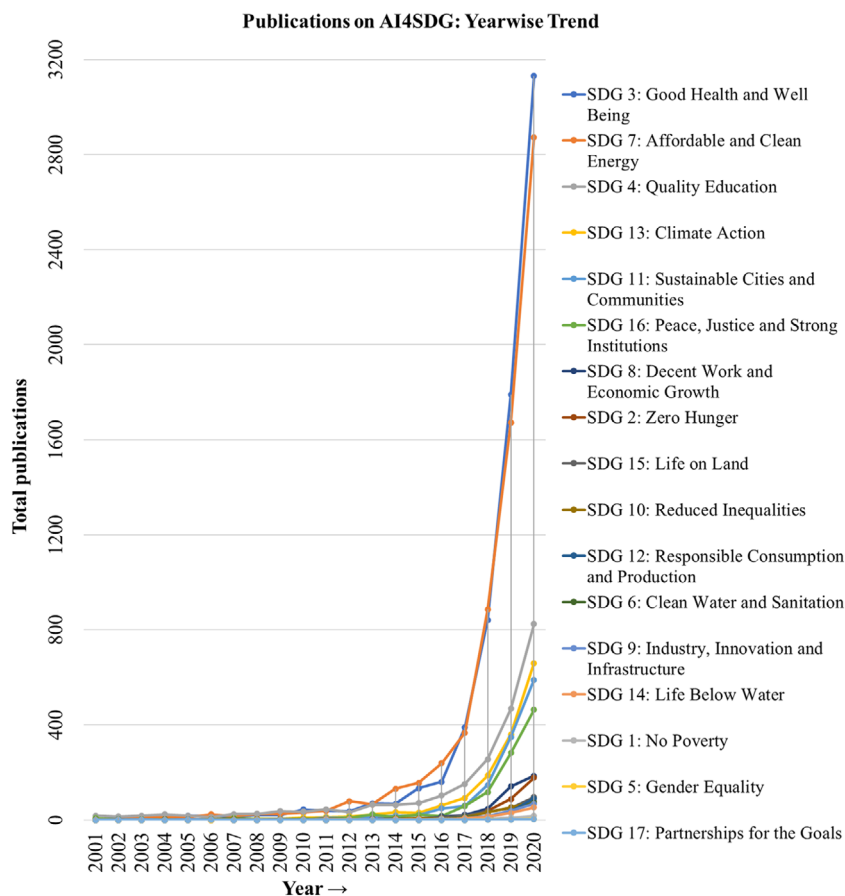
The 20,511 articles published were distributed in 360 journals, showing the wider applications of AI in SDGs. Among these journals, the

top 20 journals ranked by the number of publications were selected. These accounted for 17.13% (3515) of the total publications in AI4SDG. The top journal with the most publications was IEEE Access (published by the Institute of Electrical and Electronics Engineers), followed by the Journal of Physics Conference Series. The top 3 journals ranked by Citations per paper (CPP) included Applied Energy (52.38), IEEE Internet of Things Journal (29.18), and Scientific Reports (27.96). The total citations received by papers published in Applied Energy were the highest followed by IEEE Access and Sensors (Table 1). The journals that appear in the top list vary in subject classifications. Many of them were found multidisciplinary. While the other prominent subjects having publications in AI4SDGs include Computer Science, Electrical & Electronics, Chemistry, Instruments & Instrumentation, Geosciences, Civil engineering, Energy & Fuels, and Thermodynamics. This in turn portrays a multifaceted penetration of AI in different domains.

4.5 | AI research output related to various SDGs

SDGs 3 (Good Health and Well Being) and SDG 7 (Affordable and Clean Energy) showed a sharp rise in the total number of publications from 2015 onwards with the number of publications doubling from the previous year (Figure 4). SDG 3 had the highest number of total

FIGURE 4 The trend in publication count of AI applications on individual SDGs. Three groups based on the number of publications can be seen, these include SDGs 7 and 3 (High), 4, 13, 11, 16 (Moderate), and the rest 11 (Low). SDGs 7 and 3 exhibit a rapid increase since 2012. Before 2012, the highest number of publications was 45 (2011) for SDG 4. The legend lists the SDGs based on their total yearly publications. AI, artificial intelligence; SDGs, Sustainable Development Goals.



publications (6829), followed by SDG 7 (6683), SDG 4 (2312), SDG 13 (1510), SDG 11 (1296), and SDG 16 (1059). Other goals however had less than 1000 total publications on AI4SDG. The number of publications in SDG 17 (Partnerships for the Goals) was the smallest. These patterns indicate that AI has a more involved role in certain areas as compared to others.

4.6 | Region – AI4SDG publications map

The highest proportion of research publications from the United States as well as China corresponds to topics related to SDG 3 and SDG 7. Publications corresponding to topics related to SDG 1, and SDG 5 correspond to nine countries and SDG 17 corresponds to only two countries, namely, the United States and the United Kingdom. Both region-wise and SDG-wise differential focus in research relating to the application of AI in SDG can be inferred from the heatmap (Figure 5).

5 | DETAILED ANALYSIS OF PROBLEM DOMAINS ADDRESSED, AND AI TECHNIQUE APPLIED FOR SDGS

This section presents analysis of the major problem areas being addressed with the help of AI techniques among various SDGs. As

the publications count was unevenly distributed among these SDGs, only the goals having high publication counts (greater than 1000) were chosen for this exploration, that is, (SDG3, SDG 7, SDG 4, SDG 11, SDG 13, SDG 16). This filtering was adopted considering an assumption that the SDG having higher research output in the domain is well explored and would represent a better AI penetration in the said area. The following subsections present a detailed analysis along each of the six SDGs considered here. For each SDG, the subject area distribution of the research output, the most frequent concepts occurring in the research papers on that SDG, the major concept evolution paths, and some highly cited papers are discussed. Figure 6 below presents a summary diagram of subject area distribution of research output in the six SDGs considered. The subsections for each SDG will discuss the relevant part of the subject area distribution shown in the figure. Plots of frequency of occurrence of the major concepts for each of the six SDGs is given in Appendix B.

5.1 | SDG 3 – Good health and well-being

This goal aims to provide a healthy lifestyle and promote good health for people of all ages. It is also one of the most researched goals as observed in the publication data analysed with 6829 total papers (comprising about 33.3% of the whole publications).

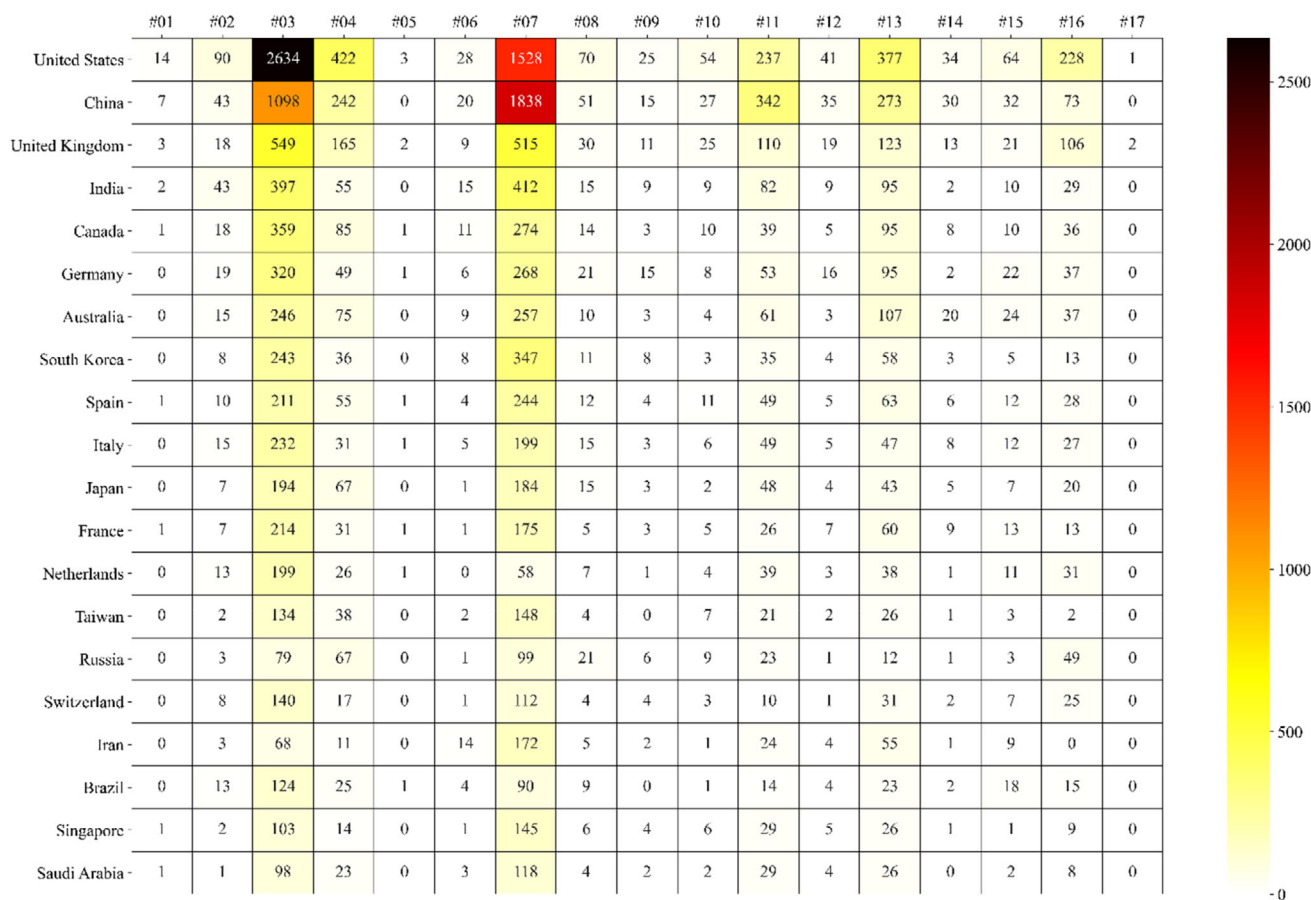


FIGURE 5 Region-wise research productivity in each SDG. Only a few countries have publications in SDG 17.

The United States was observed to be the top contributing country in the domain with ($n = 2634$) total publications. The major contributing subject areas to this SDG are found to be Medical & Health Science (37.83% papers) and Information & Computing Science (31.17% papers) (see Figure 6). The AI-related concept that topped in appearance was “convolutional neural network” with a total of 375 publications discussing it. There is a high level of employment of techniques for Image processing and Computer vision in the domain. Most of this is centred around medical diagnosis. The other concepts that appeared frequently and were directly related to the domain were “electronic health records” and “clinical decision support system”. A frequency of occurrence plot of the major concepts for SDG 3 is given in Figure A1 of the Appendix B. To get more detailed insight about the thematic attention of AI applications on this SDG, some of the most cited papers are explored in more detail next.

The most cited publication in this sub-domain was “Radiomics: the bridge between medical imaging and personalised medicine” by Lambin et al. (2017) published in “Nature Reviews Clinical Oncology” with 1366 citations. The paper talked about the importance of Radiomics, a field that describes the mining of quantitative features from medical images and coupling it with the clinical decision support system. The

paper compared the current models in practice with the Radiomics model to prove its significance and opportunities that may arrive in the future and also talked about the need for standardisation in data collection, evaluation criteria, and reporting guidelines in the field. Another paper with more than 1000 citations was from van Griethuysen et al. (2017) titled “Computational Radiomics System to Decode the Radiographic Phenotype”. They have developed an open-source platform “PyRadiomics” intended to extract a large panel of engineered features from the medical image. Another publication with significant attention was titled “Artificial Intelligence Distinguishes COVID-19 from Community-Acquired Pneumonia on Chest CT” by Li et al. (2020) and proposed a deep learning-based model to identify COVID-19 on chest CT exams and distinguish it from community-acquired pneumonia. “An immunogenic personal neoantigen vaccine for patients with melanoma” a paper from Rajkomar et al. (2018) criticised the use of standard statistical analysis on EHR as they tend to miss many relevant inferences and proposed a deep learning-based method to handle the same efficiently. Thus, AI applications in medical diagnostics, electronic health records, personalised medicine are some of the focus areas in this SDG.

Now we move ahead to present the knowledge flow and content analysis of papers in AI applications to SDG 3 through the Path analysis approach.

5.1.1 | Trends of knowledge flow in AI4SDG research related to SDG 3

SDG3 network consists of 5625 nodes and 5509 links. There is a single prominent chain that is found to be the most relevant for SDG3, **Xiong et al. (2014)**–**Lutz et al. (2020)** (Figure 7). There is a point of knowledge convergence at **Menke (2018)** and divergence at **Kessler et al. (2019)**.

Personalised medicine was the area in which the articles featured on the most prominent path were mostly focused upon. Its variations from genetics, pharmaco-genomics, and pharmacotherapy are seen among the application areas (Table 2).

5.2 | SDG 7 – Affordable and clean energy

The bigger picture of this goal is to ensure modern, clean, sustainable, and affordable energy available to all. With ($n = 6683$) number of total publications, this goal is second in terms of number of publications on the use of AI technologies. The most output in this area was found to be from China with 1838 total publications followed by US and UK. Figure 6 shows the subject areas which contribute to research on AI applications in SDG 7. The major contributing subject areas to this SDG are found to be Engineering (32.97% papers) and Information & Computing Science (31.92% papers). Next, the major concepts occurring in the research publications are shown in Figure A2 in the

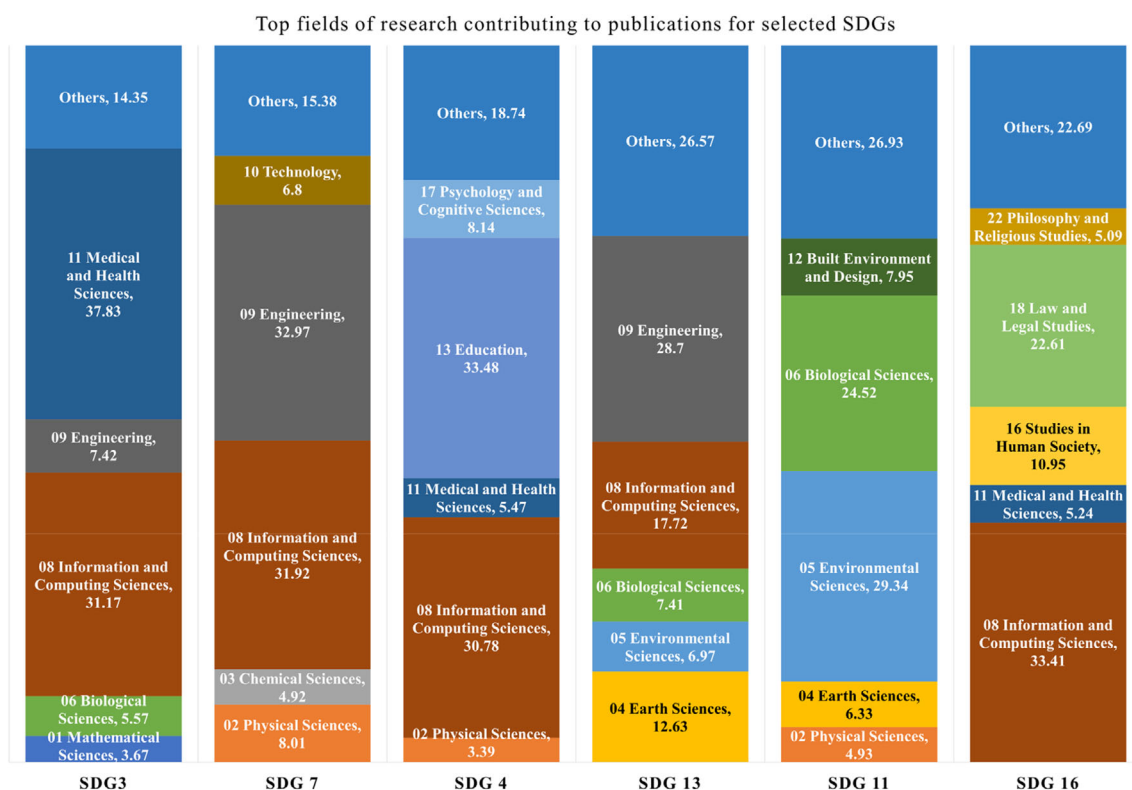


FIGURE 6 Top fields of research contributing to publications for the six SDGs considered. The percentage contribution of each field is shown along-with the classification code for the field in Dimensions data.

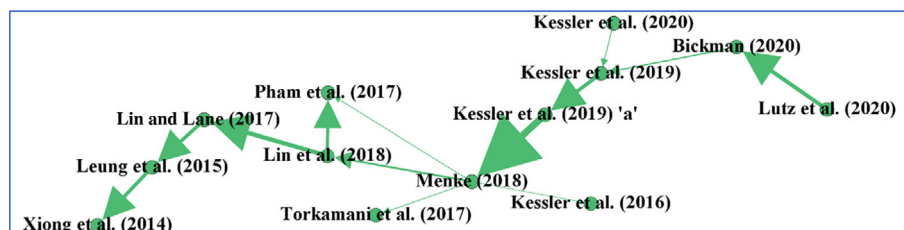


FIGURE 7 Most prominent path of knowledge flow in the citation network of publications on SDG3. In this and the following figures, the nodes represent the articles and the arrows represent the citation relationship between the articles. The size of the arrows indicates FV normalised count of search paths passing through the pair of nodes.

TABLE 2 Year-wise listing of major areas of application and targets in SDG3, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	Theme	SDG 3 target covered	AI method(s)/Models studied/used	Nature of the study	Inference/Conclusion
2014	Genetics Research	Precision Medicine/ Genomic Medicine	3.D	Bayesian deep learning algorithm	Predictive Analysis	A model for identification of genes causing disease
2015	Genetics Research			Deep learning, Recurrent Neural Networks	Review	Computational Model of Protein-DNA and Protein-RNA Binding
2017	Pharmaco-genomics (Systems genomics)	Systems genomics		Machine learning and systems genomics (MLSG)	Review	MLSG tools-based prediction of diseases and treatments
2018	Use of Genetic Biomarkers for Clinical Decision Making	Personalised Medicine/ Genomic Medicine		Multilayer feedforward neural networks (MFNNs)	Predictive Analysis	Deep MFNN framework-based tool for distinguishing treatment responders
	Pharmacotherapy	Precision Medicine/ Personalised Medicine		Machine learning and Deep learning	Review	Stratification of patients allows for tailored treatments, increasing response rates, and reduce medical errors.
2019	Psychology	Precision Healthcare	3.4.		Descriptive Analysis	Different scenarios for use of ML methods in mental healthcare
				Ensemble Learning Models using Super learner	Predictive Analysis	
2020			3.4. 3.D		Descriptive Analysis	Contributions of AI in improving mental health services
					Descriptive Analysis	Development of mental health services as a whole

Appendix B, in terms of a frequency plot of the major concepts occurring in the research papers on this SDG. “Neural networks” was the most iterated technological concept with 525 appearances followed by “convolutional neural networks” with more than 300 appearances in various publications. Frequently occurring concepts related to the problem domain were “wireless sensor networks”, “smart grids” and “electric vehicles”.

Now we look at some most cited papers on AI applications on SDG 7. One of the most cited publications (more than 1200 citations) in the domain was “Fog and IoT: An Overview of Research Opportunities” by Chiang and Zhang (2016) which is a survey exploring the opportunities generated by the development of Fog computing and IoT in the areas such as car assembly plants, minimising the downtime of nuclear reactors, optimising electrical power generators in smart energy grids. Another paper titled “A review on the prediction of building energy consumption” was by Zhao and Magoulès (2012) that appeared in the “Journal of Renewable and Sustainable Energy Reviews” surveyed the recent works that have explored the probable solutions to the problem of predicting the energy consumption of a buildings considering various sub-factors like ambient weather conditions, building structure, and so forth. Another survey paper from Foley et al. (2012) titled “Current methods and advances in the

forecasting of wind power generation” discussed various methods and techniques available to predict wind power generation considering the climatic and other factors into consideration, and got significant attention. Another publication on a similar theme was “Machine learning methods for solar radiation forecasting: A review” by Voyant et al. (2017), published in the journal “Renewable Energy” and it compared the various popular machine learning-based methods in predicting the power output of the solar system. ANN and ARIMA-based methods were their major focusing area. AI applications in management of energy systems is a major focus area in this SDG.

5.2.1 | Trends of knowledge flow in AI4SDG research related to SDG 7

The path analysis approach shows that the SDG7 network consists of 6084 nodes and 7522 links. Figure 8 shows that the obtained network path had various chains and multiple points of divergence, indicating the highly interdisciplinary nature of research in SDG7. Three of the longest chains started from Ilagras (2008), Paris et al. (2011), and Chaouachi et al. (2010) and terminated at Yoon et al. (2020), Vaisakh and Jayabarathi (2020), and Tovar et al. (2020) respectively.

There are two points of knowledge convergence **Mocanu et al. (2016)** and **Reynolds et al. (2018)** while three points of divergence **Ahmad et al. (2018)a**, **Ghimire et al. (2019)**, and **Wang et al. (2020)**.

The detailed content analysis of the papers shows that the articles focusing on using AI and technology to promote Energy conservation, Energy Generation, Solar power generation, and so forth, are found to be present on the most prominent path (Table 3).

5.3 | SDG 4 – Quality education

This SDG targets to provide completely free, equitable, and quality education for both boys and girls by 2030 so they the acquire skills and knowledge required for their sustainable development and eliminate any kind of gender biases in education. It ranked third in our list with ($n = 2312$) publications in total. Once again, United Nations appears to be the country with most publications followed by China and UK. The major contributing subject areas to this SDG are found to be Education (33.48% papers) and Information & Computing Science (30.78% papers) (see Figure 6). The major concepts occurring in the research papers in this domain are presented in Figure A3 in Appendix B. The technological concepts that dominated this area were “Artificial Intelligence” and “Natural Language Processing”. Also “Learning environment” and “Higher Education” were the most repeated problem area concept depicting the creation of an augmented learning approach with the help of AI.

Now we look at some top cited papers in this domain. A publication with significant attention in the domain was “Auto Tutor: A tutor with dialogue in natural language” by Graesser et al. (2004) which appeared in the “Behavior Research Methods” journal. This paper described “Auto Tutor”, a Natural Language Dialogue tutor which aims to assist the user to better frame their answers through a series of dialogues with Auto Tutor. Another paper titled “Learning by Teaching: A new agent

paradigm for educational software”, published in the “Applied Artificial Intelligence” journal, discussed agent-based (Betty's brain) interactive module proposed by Biswas et al. (2005). Another highly cited publication titled “Developing serious games for cultural heritage: a state-of-the-art review” by Anderson et al. (2010) discussed the use of virtual and augmented reality games to support historical teaching and learning. One more highly cited paper in this domain was “Machine Learning Paradigms for Speech Recognition: An Overview” by Deng and Li (2013) that discussed the challenges and opportunities that Artificial Intelligence poses in higher education. Applications of AI in educational technologies and paradigms is a major focus area in this SDG.

5.3.1 | Trends of knowledge flow in AI4SDG research related to SDG 4

The path analysis of papers in this domain show that SDG4 network consists of 1590 nodes and 437 links. There is a single prominent chain that is found to be the most relevant for SDG4 (Figure 9), **Hussain et al. (2018)–Zhou and Song (2020)**. There is one point of divergence **Naseer et al. (2020)** ‘a’, which links to **Naseer et al. (2020)** and **Zhou and Song (2020)**.

In the most prominent path for SDG 4, studies using Machine and Deep Learning methods in predicting performance of individuals (students, employees, teams) are found. These studies aim to assist decision making and improve the efficiency of management and enhance learning for students (Table 4).

5.4 | SDG 13 – Climate action

Mitigating adverse effects caused by climate change and improving education and awareness regarding appropriate actions towards

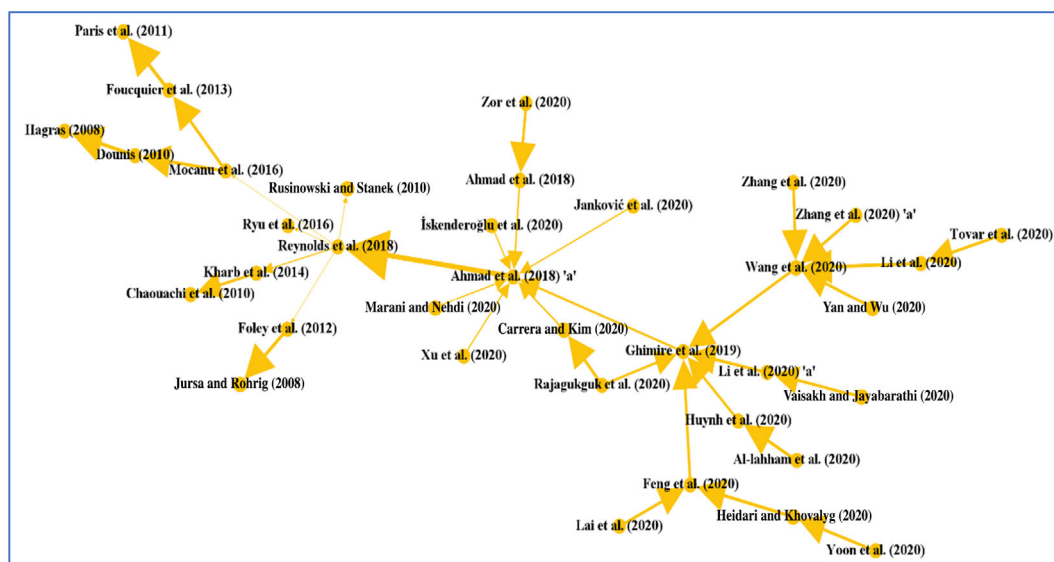


FIGURE 8 Most prominent path of knowledge flow in the citation network of publications on SDG7.

TABLE 3 Year-wise listing of major areas of application and targets in SDG7, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	Themes	SDG 7 target covered	AI method(s)/ Models studied/ used	Nature of the study	Inference/Conclusion
2008	Energy Conservation/ Internet of Things	Energy Efficient Houses/Spaces	7.3	Computational Intelligence/ Soft Computing	Descriptive Analysis	Smart devices with AI aggregation system to provide energy efficient, intelligent, adaptive and convenient user experience
2010				Computational Intelligence/ Soft Computing, Distributed AI/ Ambient AI	Descriptive Analysis	Intelligent Agents and Multiagent Systems in energy conservation in Buildings
2016	Energy Conservation	Smart Grids	7.A	Artificial Neural Network, Deep learning, CRBM, FRBM, SVM, RNN	Predictive Model Analysis	FCRBM's advantage over other methods for predication of building energy consumption patterns
2018	Energy System Optimization	Energy Generation and Optimisation		Deep learning, AI for district energy conversion technology	Review	Proposal for a model future district energy management solution using semantic modelling
2019	Energy System Optimization	Energy Efficient Houses/Spaces	7.3	DHN, Extremely Randomised Tree (ET), Support Vector Regression (SVR)	Predictive Model Analysis	All DHN, ET and SVR are good for predicting HVAC hourly energy consumption
2020	Solar Power Generation	Energy Generation and Optimisation	7.2	Convolutional Neural Network (CNN), Long Short-Term Memory Network (LSTM)	Predictive Model Analysis	A hybrid deep learning model to forecast global solar radiation
2020	Solar Assisted Water Heating	Energy Efficient Houses/Spaces	7.3 7.A	LSTM, Attention-based LSTM, ALSTM – decomposed data (ALSTM-D)	Validation Study	Superior performance of ALSTM-D in predicting energy use over conventional models
2020	Energy Efficiency	Building Energy Use	7.2 7.A	ANN, CNN, RNN, LSTM	Predictive Model Analysis	Higher prediction accuracy of building energy usage of LSTM over other methods
2020	Solar/PV Power Generation	AI Methods for Prediction of Solar energy availability		Multiple Methods Described	Review	Taxonomy of existing models for prediction of solar power availability and their applications
2020		PV power systems/ Smart Grid		CNN, LSTM (RNN)	Review	Hybrid Deep Learning approach for PV output power forecasting
2020		PV power systems		CNN, LSTM (RNN), Ridge Regression and Lasso Regression	Predictive Model Analysis	Hybrid Neural Network has better prediction of PV power generation for energy systems

Abbreviations: CNN, convolutional neural network; CRBM, conditional restricted Boltzmann machine; DHN, deep highway networks; ET, extremely randomised tree; FRBM, factored conditional restricted Boltzmann machine; LSTM, long short-term memory network; RNN, recurrent neural networks; SVR, support vector regression.

climate is the focus area of this SDG. With ($n = 1510$) publications in the corpus, this ranked 4th in the use of AI to address consequent problems. About 86% of the total output in this area was observed to

be published after 2015. The Paris Agreement 2015 could be considered a probable catalyst for this increase in research interest. The US, China, and the UK again appear to be the countries with higher

publications in the domain. The major contributing subject areas to this SDG are found to be Engineering (28.7% papers) and Information & Computing Science (17.72% papers) (see Figure 6). Frequently appearing concept relating to AI was observed to be “Neural Network”. While “Climate Change” and “Renewable Energy Resources” were the problem-specific popular concepts (see Figure A4 in Appendix B).

Now we look at some top cited papers in this domain. One of the most cited publications in this area “Recent decline in the global land evapotranspiration trend due to limited moisture supply” by Jung et al. (2010) published in “Nature” journal, suggested an ensemble-based machine learning method to assess the change in Evapotranspiration using geospatial data from satellite remote sensing and surface meteorological data. Another publication by Kong et al. (2017) titled “Short-Term Residential Load Forecasting Based on LSTM Recurrent Neural Network” talked about the employment of LSTM-based recurrent neural network to forecast electric load requirements at a micro level (single energy customer). One more popular paper in the field was “Multi objective Intelligent Energy Management for a Microgrid” by Chaouachi et al. (2012) published in “IEEE Transactions on Industrial Electronics”, and it proposed a smart solution for microgrid power prediction and optimisation problems with an AI-based technique together with the linear programming-based multi-objective optimisation. Palliating the world’s carbon debt is another burning issue in this area, with a publication from Sanderman et al. (2017) titled “Soil carbon debt of 12,000 years of human land use” having addressed this issue through a machine learning model on a global compilation of soil organic carbon data along with (HYDE) land use data and several other climatic data. AI applications in climate data handling and energy related systems management is a major focus area in this SDG.

5.4.1 | Trends of knowledge flow in AI4SDG research related to SDG 13

The path analysis of papers in this domain show that SDG 13 network consists of 1401 nodes and 778 links. There is a single prominent chain that was found to be the most relevant for SDG13 (Figure 10), Assouline et al. (2017)–Ayzel and Izhitskiy (2019). There is one point of divergence Bogner et al. (2019), which links to Croce et al. (2020) and Ayzel and Izhitskiy (2019).

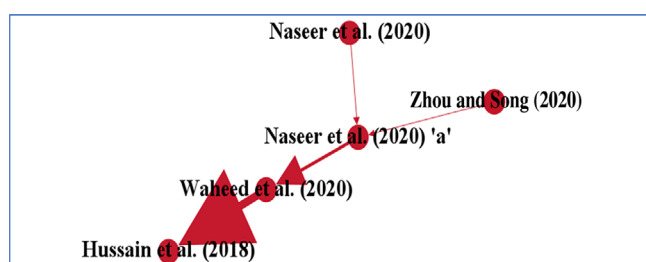


FIGURE 9 Most prominent path of knowledge flow in the citation network of publications on SDG4.

TABLE 4 Year-wise listing of major areas of application and targets in SDG4, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	SDG 4 target covered	Theme	AI method(s)/Models studied/used	Nature of the study	Inference/Conclusion
2018	Machine Learning for predictive analysis	4,A	Online Education	J48, decision tree, JRIP, and gradient-boosted classifiers	Predictive Model Analysis	Models to evaluate student engagement in online VLEs.
2020	Deep Learning for predictive analysis	4,4		deep artificial neural network, baseline logistic regression and support vector machine models		Strategies and toolkits for prediction of student performance
	Machine Learning for predictive analysis	4,A	Software Development	J48 decision tree, Naive Bayes (NB), ANN, LR, SLR, RIPPER, and SMO		a hybrid approach of information gain feature selection to predict the teams expected to obtain a below-average grade in software product development
	Machine Learning for predictive analysis		Software Development	Random forest (RF) algorithm, bagging, J48, sequential minimal optimization (SMO), multilayer perceptron (MLP), and Naive Bayes (NB)		Models to predict the Success of project teams
	Big Data		Online Education	Apriori algorithm	Recommendation Analysis	Recommendation system for course educators

Studies featured on the most prominent path for SDG 13 presented machine learning models for prediction and increasing operational efficiency of electrical systems. These studies pertain to areas of Renewable and Non-Renewable electricity generation technologies (Table 5).

5.5 | SDG 11 – Sustainable cities and communities

Making human settlements a better place, raising the standard of living of an individual, and eradicating slums are some of the prime focus areas of this SDG. The use of AI to address problems relating to this

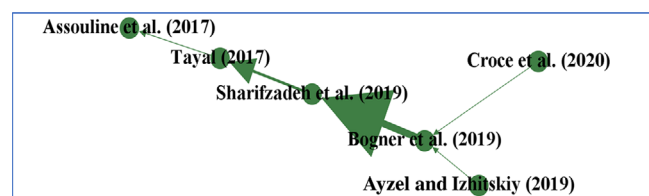


FIGURE 10 Most prominent path of knowledge flow in the citation network of publications on SDG13.

TABLE 5 Year-wise listing of major areas of application and targets in SDG 13, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	Theme	SDG 13 target covered	AI method(s)/ Models studied/used	Nature of the study	Inference/Conclusion
2017	PV Solar Energy Potential	Renewable energy	13.3 13.2	Support Vector Machines (SVM) Geographic Information Systems (GIS)	Predictive/ Validation	Estimation strategy using SVM and GIS for Rooftop solar PV
2017	Renewable Energy Adoption	Renewable energy/ Grid Management	13.3		Descriptive Study/ Review	Possible penetration of renewable energy with existing limitations of electrical grids
2019	Power Generation	Renewable energy/ Grid Management	13.3	ANN, Support Vector Regression (SVR), Gaussian process regression (GPR)	Validation	Models based on ML algorithms to predict wind and solar power potential and electricity demand for a given grid system
2019	Energy Production/ Consumption	Energy Production	13.2 13.3	Multivariate Adaptive Regression Splines (MARS), Quantile Random Forest (QRF), Gradient Boosting Machines (GBM), Nonhomogeneous Gaussian Regression (NGR)		Usefulness of ML methods in accurate estimation of energy consumption and production at the level of grids
2019	Climate Change	Hydrology/ Ecology	13.1 13.B	Extreme Gradient Boosting (XGB) Machine		Model for prediction of impact of climate change on freshwater inflow in Aral Sea
2020	Renewable Energy Adoption	Energy Use/ Smart Grid	13.3 13.B	Multiple Linear Regression	Predictive/ Validation	Methodology for accurate planning of energy supply in districts based on energy demand estimation

SDG was also found to be popular. With ($n = 1296$) publications it stands at 5th place in the list. Here, a lead of China was observed with 26.4% of publication share followed by the US and UK. The major contributing subject areas to this SDG are found to be Environment Science (29.34% papers) and Biological Science (24.52% papers) (see Figure 6). “Convolutional Neural Network” and “Internet of Things” were the most occurring technological concepts, while “Smart Cities” was the most popular problem-specific concept, showing a shift of paradigm from conventional to revolutionary technologies to address the various problems in the urban planning domain (see Figure A5 in Appendix B).

Now we look at some most cited papers in this domain. The most cited paper in the domain was “Forecasting energy consumption of multi-family residential buildings using support vector regression: Investigating the impact of temporal and spatial monitoring granularity on performance accuracy” with 326 citations by Jain et al. (2014) and was published in “Applied Energy” journal. It discussed about predicting energy consumption in a multi-family building system using regression and suggested an advanced way of development and installation of smart electricity metering devices for smart homes. Another paper discussing smart homes was “The Smart House for Older Persons and Persons with Physical Disabilities: Structure,

Technology Arrangements, and Perspectives” by Stefanov et al. (2004) appeared in the journal “IEEE Transactions on Neural Systems and Rehabilitation Engineering” and explored the various ways through which smart homes can help the aged and disabled people in the current scenario. The paper “Survey of computational intelligence as the basis to big flood management: challenges, research directions and future work” by Fotovatikhah et al. (2018) was another interesting review paper to have explored the AI and CI (computational intelligence) based approaches for the problem of the flood management. Another interesting publication with high attention was “Long short-term memory neural network for air pollutant concentration predictions: Method development and evaluation” by Li et al. (2017) published in the “Environment Pollution” journal. It proposed a novel LSTM-based model to predict the air pollutant concentration to ensure a better standard of living in big cities. AI applications in smart buildings and homes and pollution related aspects is a major focus area in this SDG.

5.5.1 | Trends of knowledge flow in AI4SDG research related to SDG 11

The path analysis of the papers in this domain show that SDG11 network consists of 1149 nodes and 798 links. Two chains can be seen in the most relevant path obtained from the citation network (Figure 11), namely, Xu et al. (2018) – Abdollahi et al. (2020) and Xu et al. (2018) – Abdollahi et al. (2020) ‘a’. It is also noteworthy that the existing research in AI4SDG related to SDG 11 is relatively recent, starting only in 2017–18.

The studies in SDG 11 were related to the areas of Building Segmentation, Extraction, Footprint estimation based in pattern recognition and using computer vision models (Table 6).

5.6 | SDG 16 – Peace, justice, and strong institutions

It is one of the most versatile goals among SDGs. It stresses on ensuring rule of law at all levels (national/international), reducing violence and deaths among all classes of social strata, warranting transparent and responsive governmental institutions, and assuring justice to all. This SDG also had a fair share of AI implications as a total of ($n = 1059$) publications appeared in the publication data. The

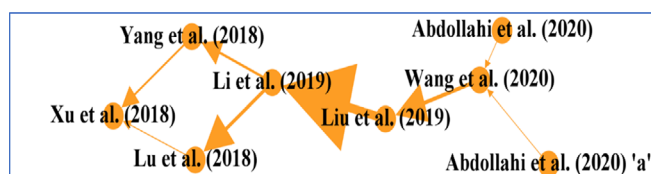


FIGURE 11 Most prominent path of knowledge flow in the citation network of publications on SDG11.

dominance of the three nations the US, UK, and China respectively was again observed here. With about 81% of the publications published after 2017, we may say that this SDG has recently started gaining attention among the AI researcher community. The major contributing subject areas to this SDG are found to be Information & Computing Science (33.41% papers) and Law & Legal Studies (22.61% papers) (see Figure 6). “Neural Network” and “Big Data” were one of the most ingeminated concepts relating to AI in the area. On the other hand, “Human Rights” and “Legal Regulations” were their domain-specific counterparts, suggesting, an emphasis on non-conventional methods to supplement the conventional ones in these sensitive areas (see Figure A6 in Appendix B).

Now we look at some of the most cited publications in this domain. The publication with the most attention (citation = 821) was “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead” by Rudin (2019) published in the journal “Nature Machine Intelligence”, and it stressed the interpretability of AI and machine learning-based algorithms for criminal justice and medicine. Another notable publication “Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities, and challenges toward responsible AI” by Arrieta et al. (2020) was a review paper in a similar discipline, that is, explainable AI and it tried to segregate different societal stakeholders in the AI paradigm and identified their challenges in trusting AI using a taxonomical approach. Suicides have become a serious issue lately in our modern society, as shown by a paper “Predicting Suicides After Psychiatric Hospitalization in US Army Soldiers: The Army Study to Assess Risk and Resilience in Service members (Army STARRS)” by Kessler et al. (2015). The paper proposes a regression tree-based ML model that was trained on data collected from various sources like sociodemographic, US Army career, criminal justice, pharmacy, and so forth, to predict the post-hospitalisation suicide possibilities in US soldiers. One more popular paper in the domain was “The accuracy, fairness, and limits of predicting recidivism” by Dressel and Farid (2018). That criticised the existing criminal risk assessment tool being used for the purpose and compared the results with a human based prediction. Explainability of AI models for tasks in this domain is a key research area in this domain.

5.6.1 | Trends of knowledge flow in AI4SDG research related to SDG 16.

The path analysis of the papers in this domain show that SDG16 network consists of 703 nodes and 251 links. The highest relevance was seen for the path shown in Figure 12. There are three chains, Berk (2009) – Bernecker et al. (2018), Berk (2009)–Rosellini et al. (2018), and Berk (2009)–Ghasemi et al. (2020). These chains consist of three nodes of divergence namely, Beck and Bleich (2013), Rosellini et al. (2015), and Street et al. (2016).

It is observed that the main areas of application of AI for SDG 16 were, Criminology, Public Health and Psychology with most of the studies being based on predictive models based on regression techniques (Table 7).

TABLE 6 Year-wise listing of major areas of application and targets in SDG11, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	Theme	SDG 11 target covered	AI method(s)/Models studied/used	Nature of the study	Inference/Conclusion
2018	Building Segmentation	Remote Sensing/Computer Vision	11.3 11.B	Deep Learning, Residual Neural Network	Validation Study	A segmentation model using CNN and guided filter with high accuracy of building identification in Remote sensing images
2018	Building Extraction			Deep Learning, Dense Attention Networks		Dense Attention Network provides an improvement on DenseNets
2018				CNN, Richer Convolution Features Networks		RCF network-based model with better F-measures than typical methods for building edge detection
2019	Building Footprint/Pattern Recognition			Deep Learning based Segmentation Methods		Hybrid U-net based semantic segmentation model using GIS map datasets along with satellite images
2019	Building Extraction			CNN, Deep Convolutional Encoder Decoder Model		Deep learning model using spatial pyramid pooling and encoder-decode with increased efficiency
2020				CNN, Fully Convolution Network		Non-local Residual U-Shape Network composed of U-Shape encoder-decoder and non-local block has improved measures over previous state of art semantic segmentation models (FCN8s, U-Net, SegNet, Deeplab3)
2020	Building Segmentation			CNN, Deep Convolution Segenet & Unet network		SegUNet – a hybrid Deep Neural Network shows improved performance in building segmentation of FCN Network
2020	Building Footprint			GAN, BConvLSTM		An end-to-end CNN model called Generative Adversarial Network (using SegNet & BConvLSTM) with better building separation accuracy.

Abbreviations: BConvLSTM, bi-directional convolutional LSTM; GAN, generative adversarial networks.

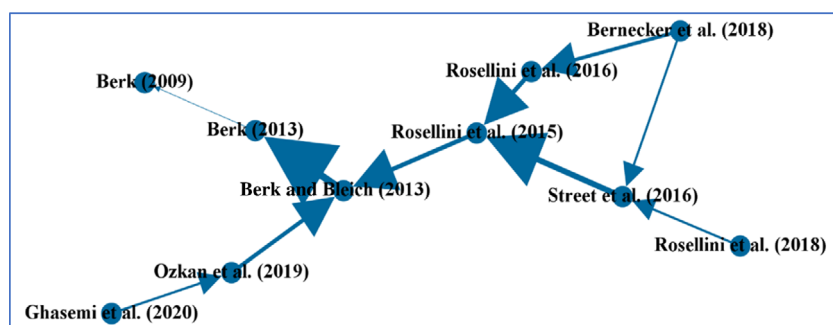


FIGURE 12 Most prominent path of knowledge flow in the citation network of publications on SDG16.

5.7 | Summarising the observed patterns in detailed analysis

The detailed analysis of articles featuring on the most prominent paths for each of the six selected SDGs provided insights on the characteristics of knowledge flow in the respective domains of research.

Tables 2, 3, 4, 5, 6 and 7 show the areas of application, themes, SDG targets covered, AI methods/models studied/used, nature of studies and inferences drawn from the studies regarding the impact of AI on the application area. Among the article featured on the most prominent paths of knowledge flow in the citation network of publications, the trends of knowledge flow are as follows.

TABLE 7 Year-wise listing of major areas of application and targets in SDG16, artificial intelligence (AI) methods and models used, nature of the study, and research findings of the available literature deduced using path analysis.

Year	Area(s) of application	Theme	SDG 16 target covered	AI method(s)/ Models studied/used	Nature of the study	Inference/Conclusion
2009	Criminology	Behaviour Forecasting	16.1 16.A	Classification and Regression Tree	Predictive Analysis	Machine learning Methods can be used to classify risk of failure of probation
2013	Criminology			Decision Tree	Descriptive Analysis	Shortcoming in modelling criminology forecasting
2013	Criminology			Classifications Trees	Predictive Analysis	Classification trees as an alternative to machine learning for forecasting criminal behaviour
2015	Public Health			Cross-Validation Stepwise Regression, Random forests, Penalised Regression		Machine learning techniques had partial success in predicting soldier's behaviour
2016	Public Health			Stepwise Regression, Penalised Regression, Random Forest		Possibility to use Machine learning based predictions in identifying high risk individuals
2018	Public Health		16.1 16.7	Discrete Time Survival Analysis		Role of different datasets in improving forecasting accuracy of Machine Learning Models
2018	Psychology		16.A	Penalised Linear Regression Model, Random Forest		Use of Machine Learning based models for individual behaviour
2018	Psychology	Mental Health	16.1 16.7	Ensemble learning using multiple ML methods		An ensemble learning model for classifying mental health risk of individuals
2019	Psychology	Behaviour Forecasting	16.2	Random Forest Classification Models		RFC based models have modest predictive accuracy for sexual offences
2020	Criminology		16.1 16.4 16.A	Decision Tree, Random Forests and SVM		Machine learning is more effective than Level of Service/ Case Management Inventory methods for predicting criminal behaviour

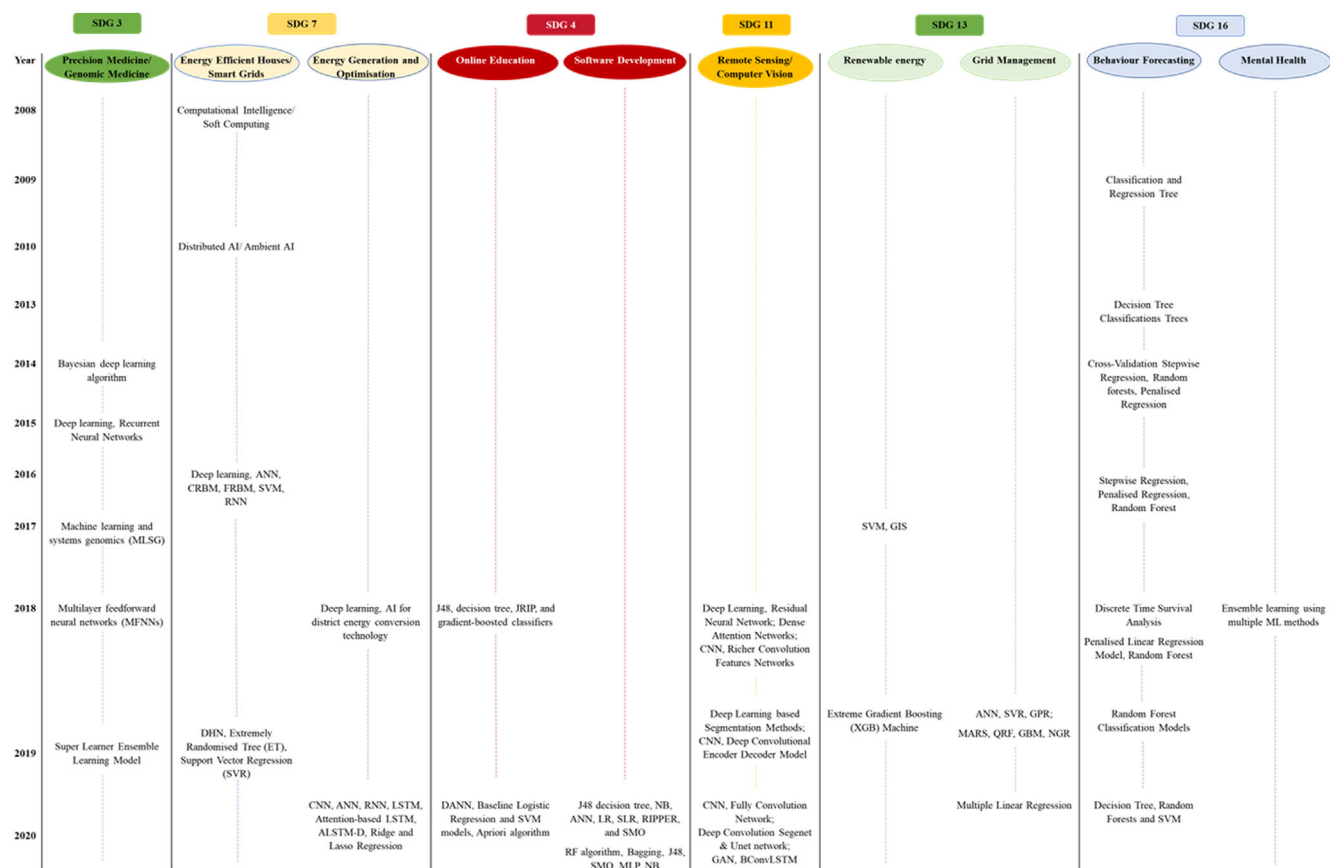


FIGURE 13 Thematic areas of research and the AI methods used in studies focused on SDGs 3, 4, 7, 11, 13 and 16 for articles identified through path analysis. AI, artificial intelligence; SDGs, Sustainable Development Goals.

1. SDG 3 consisted of Genetics, Pharmaco-genomics, Psychology related papers using Deep Learning, and Ensemble Learning methods of AI (Figure 13).
2. SDG 7 consisted of Energy Efficient Houses/ Smart Electricity Grids, and Energy generation and optimisation related papers using Soft Computing, Distributed AI, Deep Learning, and some hybrid approaches combining deep learning and machine learning (Figure 13).
3. SDG 4 has papers related to Online Education and project team efficiencies in Software Development industry. These studies focused on predictive and recommendation analysis using models such as J48, SVM, and Neural Networks (Figure 13).
4. Most prominent paths in case of SDG 11 consist of papers related to Remote Sensing and Computer vision. These utilised specific computer vision techniques for image processing, such as GAN, CNN and its variants. Hybrid approaches such as Bi-directional Convolutional LSTM are also used in recent publications (Figure 13).
5. In case of SDG 13, prominent paths have papers related to Renewable energy generation and Efficient electricity grid management approaches. Predictive analysis using regression techniques like SVR, Multiple linear regression, and ensemble approaches namely, GBM and XGB (Figure 13).
6. SDG 16 had studies mainly focusing on behaviour forecasting and mental health of convicts, soldiers, and victims of trauma. These

studies utilised regression-based Machine learning techniques such as Decision tree, Random Forest, SVM, and ML based ensemble methods (Figure 13).

For an alternative visualisation of the findings of the path analysis and subsequent content analysis of the literature on AI4SDG, Table 8 presents the list of major application areas, top keywords, top three country collaborations and the SDG targets addressed by the articles featured on the most prominent paths in path analysis.

It was found that the overall number of publications on AI4SDG has increased all across the world, with a major focus on six SDGs (3, 4, 7, 11, 13, & 16). Two of these, SDG 3 (Good Health & Well-being) and SDG 7 (Affordable and Clean Energy) were found as the areas with most applications of AI. Simple machine learning and data analytics techniques such as Decision Tree, Random Forest, SVM, SVR, Multiple linear regression, and so forth, were widely used in some of the most influential studies. Some studies also used recommendation analysis using models such as J48, SVM, and so forth. In addition, more intricate deep learning techniques like Neural Networks and Computer Vision, such as GAN, CNN, ANN and their variants are now being applied in AI4SDG. Some hybrid approaches combining deep learning and machine learning (such as SegUNet, Bi-directional Convolutional LSTM, etc.), ensemble approaches (GBM

TABLE 8 Trends in applied research and knowledge flow in top 6 Sustainable Development Goals (SDGs).

SDG	Application areas	Top keyword	Top country collaborations	SDG targets addressed (No)*
SDG 3 – Good Health and Well Being	Precision/Genomic Medicine	Convolutional Neural Networks	US-China US-UK US-Canada	3.4, 3.D (2)
SDG 7 – Affordable and Clean Energy	Energy Efficient Houses/ Smart Grids, Energy Generation and Optimisation	Neural Network	China-US China-Aus Aus-US	7.2, 7.3, 7.A (3)
SDG 4 – Quality Education	Online Education, Software Development	Learning Approach	US-Canada US-China US-UK	4.4, 4.A (2)
SDG 13 – Climate Action	Renewable Energy, Grid Management	Climate Change	US-China US-UK US-Canada	13.1, 13.2, 13.3, 13.B (4)
SDG 11 – Sustainable Cities and Communities	Remote Sensing/ Computer Vision	Smart Cities	China-US China-UK China-Aus	11.3, 11.B (2)
SDG 16 – Peace, Justice, and Strong Institutions	Behaviour Forecasting, Mental Health	Artificial Intelligence	US-UK US-China US-Germany	16.1, 16.2, 16.4, 16.7, 16.A (5)

Note: *A list of the SDG targets featured in this study is provided in Appendix C.

and XGB, etc.) have also been applied by researchers in recent years. In addition, a variation in applied techniques was also seen based on the area of application.

6 | PRACTICAL IMPLICATIONS/USEFULNESS OF THE STUDY RESULTS

The findings of the study present a comprehensive picture of the research and application landscape of AI with respect to the SDGs. Some of the practical uses/ implications of the results are as follows.

1. Specific areas of SDGs where AI can be applied are identified. Further, specific methods/models applied in these areas are also identified. This is essential information which can guide regional, national and international AI strategies.
2. Differential focus of AI on SDGs and region-specific patterns are identified which can be used to institute appropriate action plans for utilising AI technologies in the domain of SDGs in the given national/regional context.
3. Major publication venues where AI research focused on SDGs is published are identified, this can help researchers, policymakers and practitioners in benefiting from and also contributing to the existing body of knowledge.

7 | CONCLUSION

This study presents an in-depth analysis of the available literature in the area of AI4SDG. The path-analysis and content-analysis sections highlight the important areas of activity and identify the most

prominent literature in the respective areas. This work quantifies the total research activity on AI for SDGs from different parts of the world, analyse how has it changed over time, identifies which regions of the world are working towards AI applications to SDGs and how do they collaborate with each other for this purpose (RQ1 and RQ2). Our results reflect an incremental trend in overall publications on the application of AI for SDGs across the different regions of the world. SDGs 3 (good health & well-being) and 7 (affordable and clean energy) are found as the areas with the most applications of AI. SDG 4 (quality education), SDG 13 (climate action), SDG 11 (sustainable cities and communities) and SDG 16 (peace, justice and strong institutions) are the other SDGs where AI approaches and techniques are applied. In addition, the study has analysed the knowledge flows in AI research for SDGs and has identified major application areas along with AI methods and models applied to address different SDG targets (RQ3 and RQ4).

These observations can form the basis for researchers, and universities to invest in the upcoming or neglected areas of SDG research, including application of AI for the purpose. Governments can realign their policies and programmes to boost certain activities based on their national and regional preferences. Some of the observations specially from the content-analysis section are subjective and may vary in future studies depending on the analytical framework chosen by the researchers, but they provide valuable insights into the growth and development of the discipline. Alternative methods for conducting these analyses can be explored in future studies. Researchers can focus on domain specific studies to provide microscopic picture of the progress in selected domains. Regional differences in trends of research are another area where similar studies can be directed to prepare country/region specific recommendations for policy and governance interventions (Goh & Vinuesa, 2021) for accelerating progress towards the achievement of the SDGs.

ACKNOWLEDGEMENTS

The authors would like to acknowledge the support in form of the extramural research grant no. MTR/2020/000625 from Science and Engineering Research Board (SERB), India, and by HPE Aruba Centre for Research in Information Systems at BHU (No. M-22-69 of BHU).

CONFLICT OF INTEREST STATEMENT

The authors declare that the manuscript complies with ethical standards of the journal and there is no conflict of interests whatsoever.

ORCID

Anurag Kanaujia  <https://orcid.org/0000-0002-5813-7427>

Vivek Kumar Singh  <https://orcid.org/0000-0002-7348-6545>

REFERENCES

- Allen, C., Reid, M., Thwaites, J., Glover, R., & Kestin, T. (2020). Assessing national progress and priorities for the Sustainable Development Goals (SDGs): Experience from Australia. *Sustainability Science*, 15(2), 521–538.
- Anderson, E. F., McLoughlin, L., Liarakapis, F., Peters, C., Petridis, P., & De Freitas, S. (2010). Developing serious games for cultural heritage: A state-of-the-art review. *Virtual Reality*, 14(4), 255–275.
- Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bannetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., & Chatila, R. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.
- Batagelj, V. (2003). Efficient algorithms for citation network analysis, arXiv preprint, arXiv:cs/0309023.
- Baum, K., Bryson, J., Dignum, F., Dignum, V., Grobelnik, M., Hoos, H., Irgens, M., Lukowicz, P., Muller, C., Rossi, F., & Shawe-Taylor, J. (2023). From fear to action: AI governance and opportunities for all. *Frontiers in Computer Science*, 5, 49. <https://doi.org/10.3389/fcomp.2023.1210421>
- Biswas, G., Leelawong, K., Schwartz, D., Vye, N., & The Teachable Agents Group at Vanderbilt. (2005). Learning by teaching: A new agent paradigm for educational software. *Applied Artificial Intelligence*, 19(3–4), 363–392.
- Borgatti, S. P. (2005). Centrality and network flow. *Social Networks*, 27(1), 55–71.
- Brughmans, T. (2013). Networks of networks: A citation network analysis of the adoption, use, and adaptation of formal network techniques in archaeology. *Literary and Linguistic Computing*, 28(4), 538–562.
- Chaouachi, A., Kamel, R. M., Andouls, R., & Nagasaka, K. (2012). Multiobjective intelligent energy management for a microgrid. *IEEE Transactions on Industrial Electronics*, 60(4), 1688–1699.
- Chiang, M., & Zhang, T. (2016). Fog and IoT: An overview of research opportunities. *IEEE Internet of Things Journal*, 3(6), 854–864.
- Chui, M., Harryson, M., Manyika, J., Roberts, R., Chung, R., van Heteren, A., & Nel, P. (2018). Notes from the AI frontier: Applying AI for social good. *McKinsey Global Institute*, 3.
- De Solla Price, D. J. (1965). Networks of scientific papers: The pattern of bibliographic references indicates the nature of the scientific research front. *Science*, 149(3683), 510–515.
- Deng, L., & Li, X. (2013). Machine learning paradigms for speech recognition: An overview. *IEEE Transactions on Audio, Speech and Language Processing*, 21(5), 1060–1089.
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., & Lim, W. M. (2021). How to conduct a bibliometric analysis: An overview and guidelines. *Journal of Business Research*, 133, 285–296.
- Dressel, J., & Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. *Science Advances*, 4(1), eaao5580.
- Farrukh, M., Meng, F., Wu, Y., & Nawaz, K. (2020). Twenty-eight years of business strategy and the environment research: A bibliometric analysis. *Business Strategy and the Environment*, 29(6), 2572–2582.
- Foley, A. M., Leahy, P. G., Marvuglia, A., & McKeogh, E. J. (2012). Current methods and advances in forecasting of wind power generation. *Renewable Energy*, 37(1), 1–8.
- Fotovatikhah, F., Herrera, M., Shamshirband, S., Chau, K. W., Faizollahzadeh Ardabili, S., & Piran, M. J. (2018). Survey of computational intelligence as basis to big flood management: Challenges, research directions and future work. *Engineering Applications of Computational Fluid Mechanics*, 12(1), 411–437.
- Garfield, E., Sher, I. H., & Torpie, R. J. (1964). *The use of citation data in writing the history of science*. Institute for Scientific Information Inc Philadelphia PA.
- Goh, H. H., & Vinuesa, R. (2021). Regulating artificial-intelligence applications to achieve the Sustainable Development Goals. *Discover Sustainability*, 2(52), 1–6. <https://doi.org/10.1007/s43621-021-00064-5>
- Goralski, M. A., & Tan, T. K. (2020). Artificial intelligence and sustainable development. *The International Journal of Management Education*, 18(1), 100330.
- Graesser, A. C., Lu, S., Jackson, G. T., Mitchell, H. H., Ventura, M., Olney, A., & Louwerse, M. M. (2004). AutoTutor: A tutor with dialogue in natural language. *Behavior Research Methods, Instruments, & Computers*, 36(2), 180–192.
- Gupta, S., Langhans, S. D., Domisch, S., Fuso-Nerini, F., Felländer, A., Battaglini, M., Tegmark, M., & Vinuesa, R. (2021). Assessing whether artificial intelligence is an enabler or an inhibitor of sustainability at indicator level. *Transportation Engineering*, 4, 100064.
- Haenlein, M., & Kaplan, A. (2019). A brief history of artificial intelligence: On the past, present, and future of artificial intelligence. *California Management Review*, 61(4), 5–14.
- Hummon, N. P., & Dereian, P. (1989). Connectivity in a citation network: The development of DNA theory. *Social Networks*, 11(1), 39–63.
- Jain, R. K., Smith, K. M., Culligan, P. J., & Taylor, J. E. (2014). Forecasting energy consumption of multi-family residential buildings using support vector regression: Investigating the impact of temporal and spatial monitoring granularity on performance accuracy. *Applied Energy*, 123, 168–178.
- Jo, S. J., Jeung, C. W., Park, S., & Yoon, H. J. (2009). Who is citing whom: Citation network analysis among HRD publications from 1990 to 2007. *Human Resource Development Quarterly*, 20(4), 503–537.
- Jung, M., Reichstein, M., Ciais, P., Seneviratne, S. I., Sheffield, J., Goulden, M. L., Bonan, G., Cescatti, A., Chen, J., De Jeu, R., & Dolman, A. J. (2010). Recent decline in the global land evapotranspiration trend due to limited moisture supply. *Nature*, 467(7318), 951–954.
- Kessler, R. C., Warner, C. H., Ivany, C., Petukhova, M. V., Rose, S., Bromet, E. J., Brown, M., Cai, T., Colpe, L. J., Cox, K. L., & Fullerton, C. S. (2015). Predicting suicides after psychiatric hospitalization in US Army soldiers: The Army study to assess risk and resilience in servicemembers (Army STARRS). *JAMA Psychiatry*, 72(1), 49–57.
- Kong, W., Dong, Z. Y., Jia, Y., Hill, D. J., Xu, Y., & Zhang, Y. (2017). Short-term residential load forecasting based on LSTM recurrent neural network. *IEEE Transactions on Smart Grid*, 10(1), 841–851.
- Kumar, S., Sureka, R., Lim, W. M., Kumar Mangla, S., & Goyal, N. (2021). What do we know about business strategy and environmental research? Insights from business strategy and the environment. *Business Strategy and the Environment*, 30(8), 3454–3469.
- Lambin, P., Leijenaar, R. T., Deist, T. M., Peerlings, J., De Jong, E. E., Van Timmeren, J., Sanduleanu, S., Larue, R. T., Even, A. J., Jochems, A., & Van Wijk, Y. (2017). Radiomics: The bridge between medical imaging and personalized medicine. *Nature Reviews. Clinical Oncology*, 14(12), 749–762.
- Lathabai, H. H., George, S., Prabhakaran, T., & Changat, M. (2018). An integrated approach to path analysis for weighted citation networks. *Scientometrics*, 117(3), 1871–1904.

- Lathabai, H. H., Prabhakaran, T., & Changat, M. (2015). Centrality and flow vergence gradient based path analysis of scientific literature: A case study of biotechnology for engineering. *Physica A: Statistical Mechanics and its Applications*, 429, 157–168.
- Lathabai, H. H., Prabhakaran, T., & Changat, M. (2017). Contextual productivity assessment of authors and journals: A network scientometric approach. *Scientometrics*, 110(2), 711–737.
- Li, L., Qin, L., Xu, Z., Yin, Y., Wang, X., Kong, B., Bai, J., Lu, Y., Fang, Z., Song, Q., & Cao, K. (2020). Artificial intelligence distinguishes COVID-19 from community acquired pneumonia on chest CT. *Radiology*, 296, E65–E71.
- Li, X., Peng, L., Yao, X., Cui, S., Hu, Y., You, C., & Chi, T. (2017). Long short-term memory neural network for air pollutant concentration predictions: Method development and evaluation. *Environmental Pollution*, 231, 997–1004.
- Liengpunsakul, S. (2021). Artificial intelligence and sustainable development in China. *The Chinese Economy*, 54(4), 235–248.
- Liu, J. S., & Lu, L. Y. (2012). An integrated approach for main path analysis: Development of the Hirsch index as an example. *Journal of the American Society for Information Science and Technology*, 63(3), 528–542.
- Liu, Y., Huang, B., Guo, H., & Liu, J. (2023). A big data approach to assess progress towards Sustainable Development Goals for cities of varying sizes. *Communications Earth & Environment*, 4(1), 66.
- Mina, A., Ramlogan, R., Tampubolon, G., & Metcalfe, J. S. (2007). Mapping evolutionary trajectories: Applications to the growth and transformation of medical knowledge. *Research Policy*, 36(5), 789–806.
- Moavenzadeh, J. (2015, October). The 4th industrial revolution: Reshaping the future of production. In *World economic forum* (p. 57). DHL Global Engineering & Manufacturing Summit.
- Muthukrishnan, N., Maleki, F., Ovens, K., Reinhold, C., Forghani, B., & Forghani, R. (2020). Brief history of artificial intelligence. *Neuroimaging Clinics*, 30(4), 393–399.
- National Strategy for Artificial Intelligence. (2018). NITI Aayog. Retrieved from <https://niti.gov.in/sites/default/files/2019-01/NationalStrategy-for-AI-Discussion-Paper.pdf>
- Prabhakaran, T., Lathabai, H. H., & Changat, M. (2015). Detection of paradigm shifts and emerging fields using scientific network: A case study of information Technology for Engineering. *Technological Forecasting and Social Change*, 91, 124–145.
- Prabhakaran, T., Lathabai, H. H., George, S., & Changat, M. (2018). Towards prediction of paradigm shifts from scientific literature. *Scientometrics*, 117(3), 1611–1644.
- Rajkomar, A., Oren, E., Chen, K., Dai, A. M., Hajaj, N., Hardt, M., Liu, P. J., Liu, X., Marcus, J., Sun, M., & Sundberg, P. (2018). Scalable and accurate deep learning with electronic health records. *NPJ Digital Medicine*, 1(1), 1–10.
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), 206–215.
- Sætra, H. S. (2021). AI in context and the Sustainable Development Goals: Factoring in the unsustainability of the sociotechnical system. *Sustainability*, 13(4), 1738.
- Sanderman, J., Hengl, T., & Fiske, G. J. (2017). Soil carbon debt of 12,000 years of human land use. *Proceedings of the National Academy of Sciences*, 114(36), 9575–9580.
- Scardoni, G., & Laudanna, C. (2012). Centralities based analysis of complex networks. In Y. Zhang (Ed.), *New frontiers in graph theory* (pp. 323–348).
- Singh, A., Kanaujia, A., & Singh, V. K. (2022). Research on Sustainable Development Goals: How has Indian scientific community responded? *Journal of Scientific & Industrial Research*, 81(11), 1147–1161.
- Singh, V. K., Singh, P., Karmakar, M., Leta, J., & Mayr, P. (2021). The journal coverage of web of science, Scopus and dimensions: A comparative analysis. *Scientometrics*, 126(6), 5113–5142.
- Stefanov, D. H., Bien, Z., & Bang, W. C. (2004). The smart house for older persons and persons with physical disabilities: Structure, technology arrangements, and perspectives. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 12(2), 228–250.
- Tampubolon, G., & Ramlogan, R. (2007). Networks and temporality in the development of a radical medical treatment. *Graduate Journal of Social Science*, 4(1), 54–77.
- Van Griethuysen, J. J., Fedorov, A., Parmar, C., Hosny, A., Aucoin, N., Narayan, V., Beets-Tan, R. G., Fillion-Robin, J. C., Pieper, S., & Aerts, H. J. (2017). Computational radiomics system to decode the radiographic phenotype. *Cancer Research*, 77(21), e104–e107.
- Vinuesa, R., Azizpour, H., Leite, I., Balaam, M., Dignum, V., Domisch, S., Felländer, A., Langhans, S. D., Tegmark, M., & Fuso Nerini, F. (2020). The role of artificial intelligence in achieving the Sustainable Development Goals. *Nature Communications*, 11(1), 1–10.
- Vinuesa, R., & Sirmacek, B. (2021). Interpretable deep-learning models to help achieve the Sustainable Development Goals. *Nature Machine Intelligence*, 3(11), 926.
- Vinuesa, R., Theodorou, A., Battaglini, M., & Dignum, V. (2020). A socio-technical framework for digital contact tracing. *Results in Engineering*, 8, 100163.
- Voyant, C., Notton, G., Kalogirou, S., Nivet, M. L., Paoli, C., Motte, F., & Fouilloy, A. (2017). Machine learning methods for solar radiation forecasting: A review. *Renewable Energy*, 105, 569–582.
- Wamba, S. F., Bawack, R. E., Guthrie, C., Queiroz, M. M., & Carillo, K. D. A. (2021). Are we preparing for a good AI society? A bibliometric review and research agenda. *Technological Forecasting and Social Change*, 164, 120482.
- Yeh, S. C., Wu, A. W., Yu, H. C., Wu, H. C., Kuo, Y. P., & Chen, P. X. (2021). Public perception of artificial intelligence and its connections to the Sustainable Development Goals. *Sustainability*, 13(16), 9165.
- Zhao, H. X., & Magoulès, F. (2012). A review on the prediction of building energy consumption. *Renewable and Sustainable Energy Reviews*, 16(6), 3586–3592.

How to cite this article: Singh, A., Kanaujia, A., Singh, V. K., & Vinuesa, R. (2024). Artificial intelligence for Sustainable Development Goals: Bibliometric patterns and concept evolution trajectories. *Sustainable Development*, 32(1), 724–754. <https://doi.org/10.1002/sd.2706>

APPENDIX A

This appendix briefly describes the search path count (SPC) and flow vergence (FV) gradient weight assignment and search schemes.

A.1. | SPC method: A short revisit

As mentioned earlier, SPC method is one of the weight-assignment methods that belong to the SPX topic (traversal-based weight assignment methods) developed by Batagelj (2003). In SPC method, identification of all sources and sinks in the citation network is the first task. Then, for each arc in the network, the number of search paths (from sources to sinks) passing through it is computed and such a count is assigned as weight of that arc.

A.2. | FV gradient method: A short revisit

Scientific citation network, a kind of information network is an interconnected structure of scholarly publications which are linked through citation links or arc of citations. These links can be treated as a representation of the flow of information from the cited work to the citing work. Most of the vertex measures if used independently fails to reflect one important property of a vertex (work) in a citation network that arises due to the flow of information through it, namely the FV. Flow of information through a paper invokes a dominance in terms of flow convergence or flow divergence, which can be termed as flow vergence by (Prabhakaran et al., 2015). Flow vergence potential is the potential of a paper to contribute to the growth of a field via the attainment or improvement of flow divergence dominance. Thus, works which are presently in knowledge flow divergence mode can be surely regarded as works of high flow vergence potential. Even though a work is not presently in flow divergence mode, certain works can be treated as of high FV potential. This happens only if that work succeeded in delivering knowledge to other works (that cites the concerned work) that are at certain level of quality which is reflected by eigenvector centrality (Borgatti, 2005). This concept was introduced by Prabhakaran et al. (2015) and an index for reflecting the flow vergence potential was also developed. Flow Vergence index or FV index of a paper i can be computed as:

$$W_{FV_i} = \frac{indeg_i - outdeg_i}{indeg_i + outdeg_i} + eig_i \quad (1)$$

where, $indeg_i$, $outdeg_i$ and eig_i are the indegree (the number of citations received), the outdegree (number of citations made) and eigenvector centrality of paper i in the network (Scardoni & Laudanna, 2012). Now, we move on to discuss the FV gradient method as the theoretical origin and rationale behind the formulation of the FV index is already covered by (Lathabai et al., 2018), (Prabhakaran et al., 2015), (Lathabai et al., 2015), (Lathabai et al., 2017), and (Prabhakaran et al., 2018).

Since a citing-cited pair is related by knowledge flow, existence of flow vergence potential for citing and cited work also implies the existence of flow vergence gradient or gradient in flow vergence potential between them. The potential difference between works i and j connected by an arc of citation from j to i can be termed as FV gradient (Lathabai et al., 2015) and it can be computed as:

$$\Delta FV_{ij} = W_{FV_i} - W_{FV_j} \quad (2)$$

The speciality of the of FV gradient or the FV potential difference is that, usually, the knowledge flow tends to appear as to have occurred from a work of high FV potential to a work of low FV potential. Thus, in most of the cases, ΔFV_{ij} takes a positive value. However, in some special cases, when newer works (citing works) tend to outperform the former ones (cited ones) due to its intellectual merit or innovative appeal to the scientific community, we can have $\Delta FV_{ij} < 0$, making knowledge flow to occur from low potential paper to a high

potential paper. This phenomenon was termed as Flow Vergence effect or FV effect and was used to detect (Lathabai et al., 2015) and predict (Prabhakaran et al., 2018) pivot papers of paradigm shift. Computation of FV gradients makes the network a signed weighted network and hence FV gradient method of weight assignment helps for the retrieval of paths that might not be highlighted through SPX methods. To ensure that the arcs with negative signs are not missed and given high priority when certain search schemes like key-route search are employed, the following transformation of eqn. (2) should be done.

$$\Delta FV_{ij}(\text{norm.}) = 1 + \frac{\max.\Delta FV - \Delta FV_{ij}}{\max.\Delta FV - \min.\Delta FV} \quad (3)$$

Where, $\max.\Delta FV$ is the highest FV gradient weight in the network and $\min.\Delta FV$ is the lowest FV gradient weight in the network.

A.3. | Search schemes

Forward Search: In forward search, among all the arcs originating from all the sources (the papers that do not cite any other paper but get at least one citation), the ones with highest weight will be selected. The target node of that arc will be made the new source and the same procedure will be repeated in a greedy fashion till sink papers (the papers that do not get cited yet but has cited at least one paper) are found. When there are ties, both the arcs will be considered.

Backward Search: In backward search, unlike the forward search, search originates from sink papers till source papers are obtained. Everything is same as that of forward search. This was introduced by (Liu & Lu, 2012).

Global search/Critical path method: In global search method, instead of selecting an initial arc and its subsequent arcs in a greedy or 'local best' fashion, for all the source-sink paths, the total weight of the path (which is equal to the sum of weights of all the arcs that forms that path) will be computed and the highest among that will be considered as the critical path.

Key-route Search (Local): In key-route search (local), instead of initiating search from source or sink (which invokes the risk of missing the highest weighted arc in the network), search commences from the terminal nodes of key-route (the highest weighted arc). From cited work of the key-route, search proceeds as that in backward search till source is obtained and from citing work of the key-route, search proceeds in the manner of forward search till sink is found. This search scheme is the key innovation of Liu-Lu approach (Liu & Lu, 2012). This search can be conducted to retrieve multiple paths with the selection of multiple key-routes. For instance, the software package PAJEK has set the option 1-10 as default choice, by which top 10 key-routes will be selected for the search. This choice can be changed by users accordingly.

Key-route Search (global): In key-route (global), instead of tracing from both the terminal nodes of key-routes till sources and sinks are

approached, a global search is initiated from terminal nodes of key-routes. That means, from all the paths reachable from citing paper and cited paper in the key-route, the ones with largest sum of weights will be chosen.

Along with these search schemes, a provision for tolerance is also proposed by (Liu & Lu, 2012) and available in PAJEK for local search schemes such as forward, backward and key-route (local). If a tolerance of 10% (0.1 is the default value set in PAJEK) is selected, at each step, instead of choosing the largest weighted arc, all the arcs that falls with-in 90% of its weight will be selected.

A.4. | Text-integration for path analysis

Once, important paths (which are subnetworks of citation network) in the concerned literature are identified, nodes of the paths (i.e., papers) should be relabelled or current vertex labels should be replaced by terms or words that best represent the theme/contribution of that paper. In Dimensions, since each published article is associated with 'concepts' that represent certain keywords that represent the relation of work with the subfield/field in which the work belongs to, the relabelling can be easily achieved by mapping the work's label/id with the concept. Also, since each concept are associated with a relevance score or score of relevance, selection of the best representative word

can also be systematically done with ease. Thus, the paths obtained after text-integrated path analysis (TPA) can be termed as 'Concept Evolution Paths'. The methodology used for TPA is given below:

Procedure for extraction of Concept evolution Paths

Input: Citation network of published articles C_i and Work-Concept/Keyword affiliations network WK .

Output: Concept evolution paths

1. From C_i , extract SPC paths and FV paths
2. For every pair of paths P_i and P_j
 - a. Compute $U_{P_i P_j} = \frac{|P_i| + |P_j| - |P_i \cap P_j|}{|P_i| + |P_j|}$, the uniqueness index ([15]) of P_i with respect to P_j and vice-versa.
 - b. (a) If $U_{P_i P_j} \geq \delta$ (desirable value is 0.65), select both (b) Otherwise,
 - i. Select path P_i if $|P_i| > |P_j|$
 - ii. Select path P_j if $|P_j| > |P_i|$
 - iii. Select both if $|P_i| = |P_j|$
3. From $W - K$ network, extract the subnetwork $W - K(P_x)$ (where P_x is the selected path at step 3) network by choosing concept with highest relevance score
4. Obtain concept evolution paths or concept citation path $K(P_x) \rightarrow K(P_x)$ using

$$K(P_x) \rightarrow K(P_x) = (W - K(P_x))^T \times P_x \times (W - K(P_x))$$

APPENDIX B

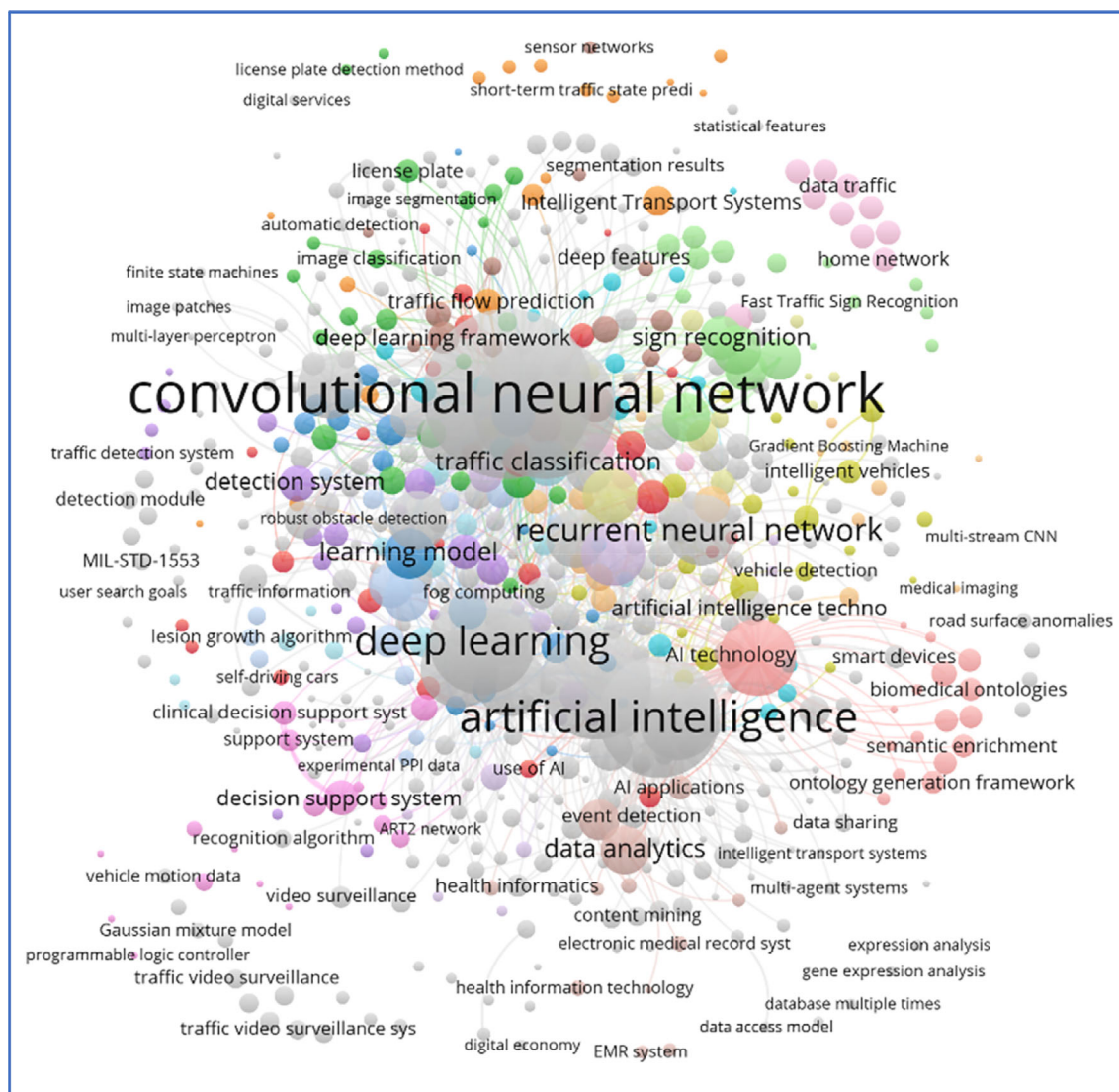


FIGURE A1 Network map of most prominent concepts from the publications on SDG3 drawn using VOS viewer. The size of nodes shows the importance of the concepts.

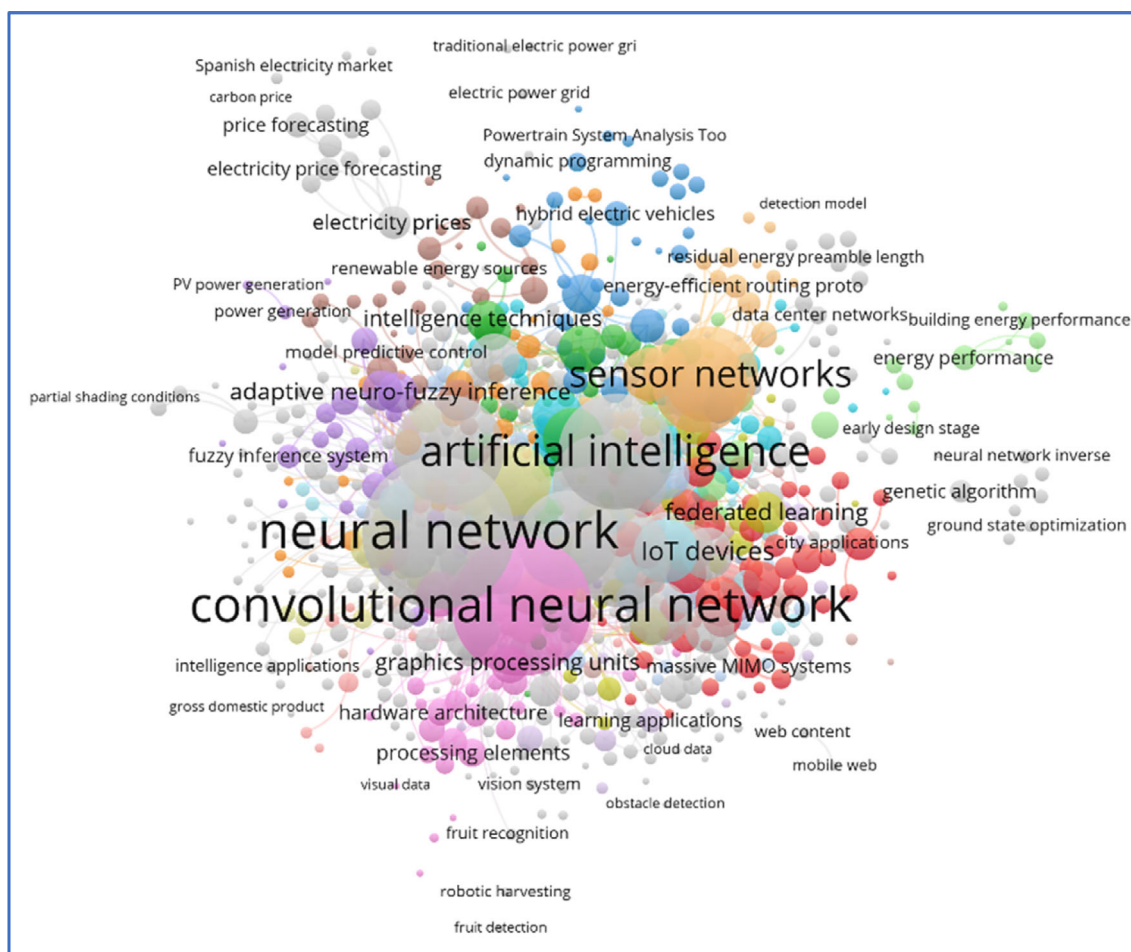


FIGURE A2 Network map of most prominent concepts from the publications on SDG7 according to their frequency and linkages. “convolutional neural network”, “neural network”, “artificial intelligence”, “sensor networks”, and “IoT devices” are the top five concepts for SDG7.

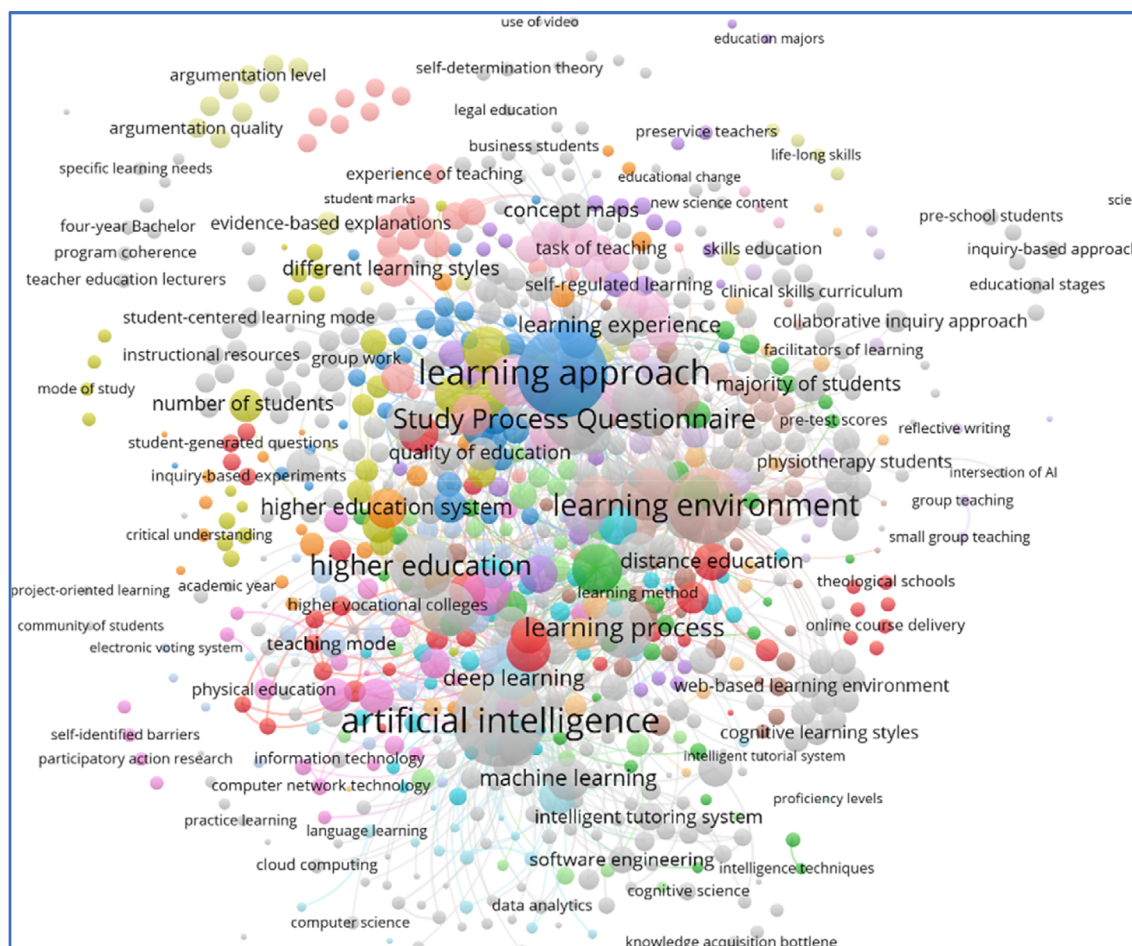
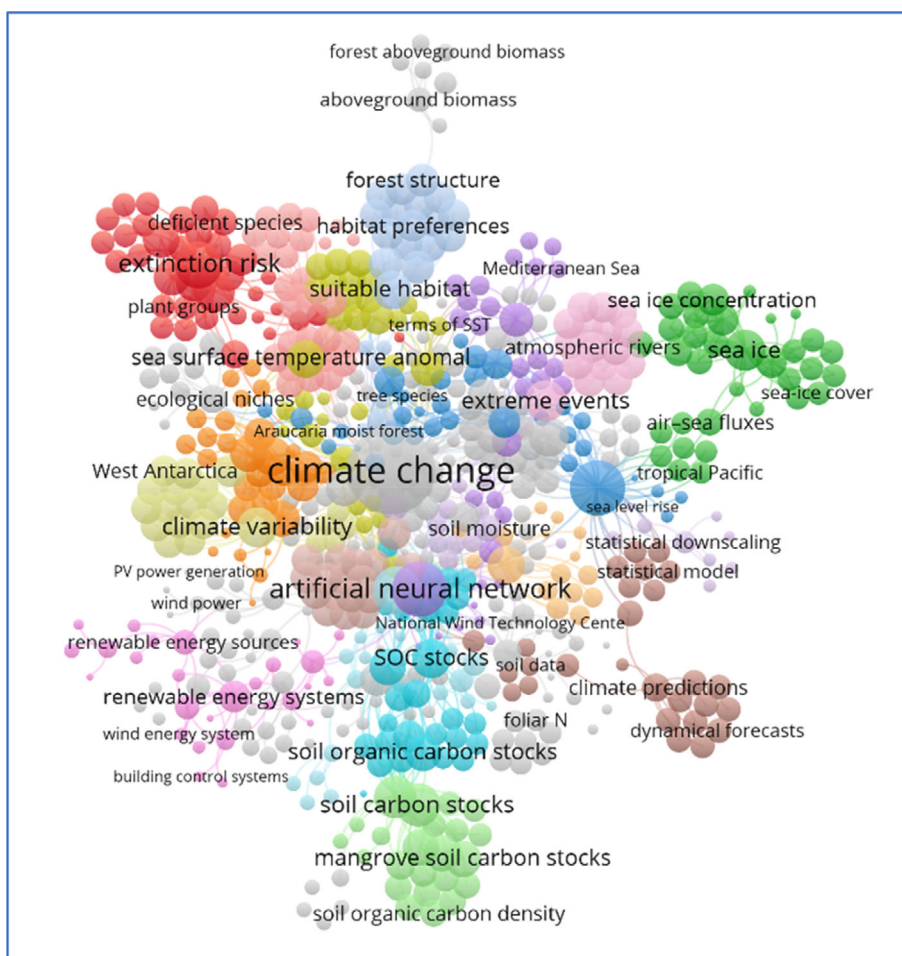


FIGURE A3 Network map of most prominent concepts from the publications on SDG4. The top five concepts for SDG4 are, “learning approach”, “artificial intelligence”, “Study Process Questionnaire”, “learning environment, and “higher education”.

FIGURE A4 Concept cluster analysis of SDG13 was conducted by drawing a co-occurrence network of concepts using Network X library (python) and plotted using VOS viewer. The most prominent concepts for SDG13 were “climate change”, “organic carbon”, “artificial neural networks”, “extinction risk” and “extreme event”.



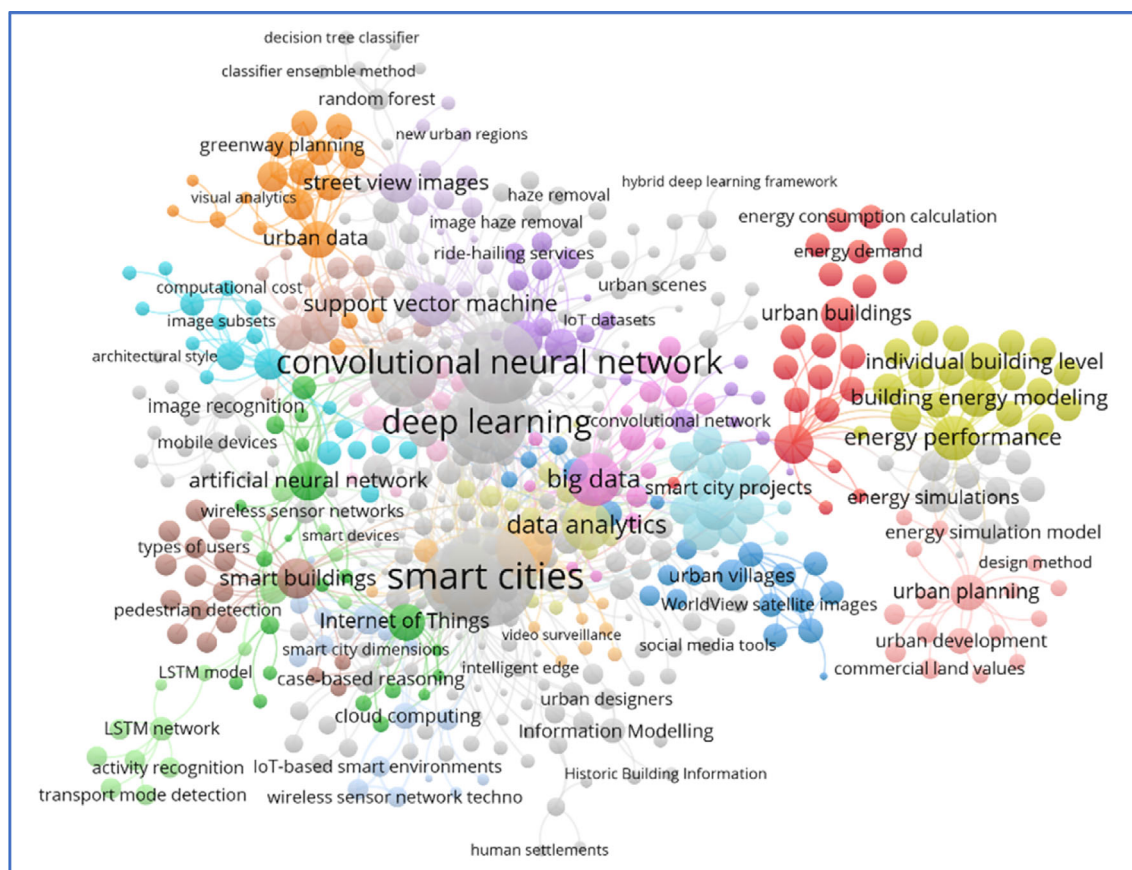


FIGURE A5 Network map of most prominent concepts from the publications on SDG11 drawn using VOS viewer. “smart cities”, “deep learning”, “convolutional neural network”, “data analytics”, and “smart buildings” are the top five occurring concepts among papers focusing on SDG11.

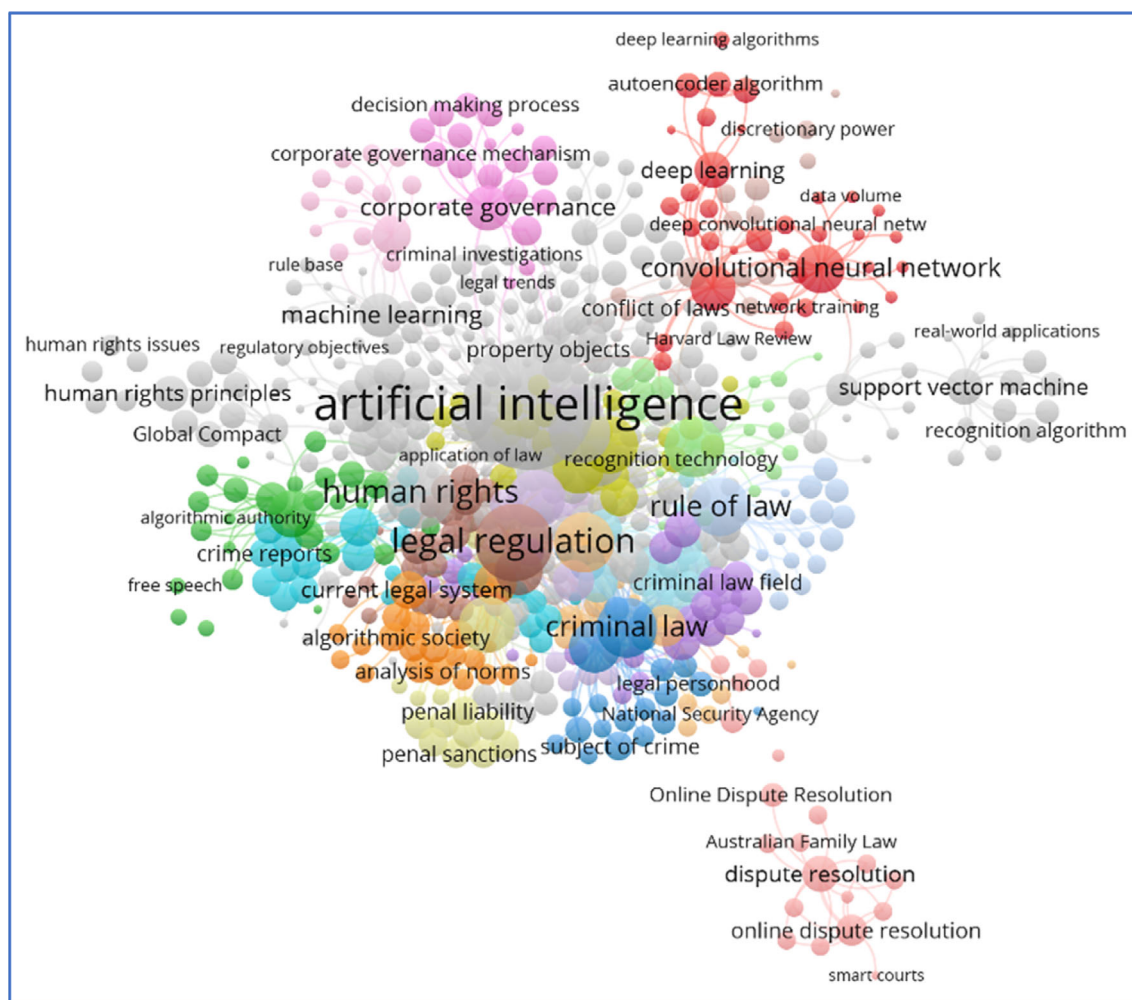


FIGURE A6 Network map of most prominent concepts from the publications on SDG16 drawn using VOS viewer. The top five concepts by frequency of occurrence are, “artificial intelligence”, “human rights”, “legal regulation”, “convolutional neural network” and “rule of law”.

APPENDIX C

SDG targets covered by studies featuring in the most prominent paths for AI4SDG research publications in selected six SDGs.

SDG targets	Target description
3.4	By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being.
3.D	Strengthen the capacity of all countries, in particular developing countries, for early warning, risk reduction and management of national and global health risks.
4.4	By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.
4.A	Build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all.
7.2	By 2030, increase substantially the share of renewable energy in the global energy mix.
7.3	By 2030, double the global rate of improvement in energy efficiency.
7.A	By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology.
11.3	By 2030, enhance inclusive and sustainable urbanisation and capacity for participatory, integrated and sustainable human settlement planning and management in all countries.
11.B	By 2020, substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, resilience to disasters, and develop and implement, in line with the Sendai Framework for Disaster Risk Reduction 2015–2030, holistic disaster risk management at all levels.
13.1	Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries.
13.2	Integrate climate change measures into national policies, strategies and planning.
13.3	Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.
13.B	Promote mechanisms for raising capacity for effective climate change-related planning and management in least developed countries, including focusing on women, youth and local and marginalised communities.
16.1	Significantly reduce all forms of violence and related death rates everywhere.
16.2	End abuse, exploitations, trafficking and all forms of violence against and torture of children.
16.4	By 2030, significantly reduce illicit financial and arms flows, strengthen the recovery and return of stolen assets and combat all forms of organised crime.
16.7	Ensure responsive, inclusive, participatory and representative decision-making at all levels.
16.A	Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime.